

1 Show that the equation $y = 16x - 17$ is a tangent to the curve $y = 3x^2 + 4x - 5$. [5]

2 If x is small and measured in radians, show that $4 \sin\left(x + \frac{\pi}{6}\right) - 8 \cos\left(x - \frac{\pi}{6}\right) = (2\sqrt{3} - 1)x^2 + (2\sqrt{3} - 4)x + 2 - 4\sqrt{3}$. [5]

3 The parametric equations of a curve are given by

$$x = 2\sin\theta + 4$$

$$y = \cos 2\theta.$$

Express the equation of the curve in terms of x and y in the form $ay = bx^2 + cx + d$ where a, b, c and d are constants. [6]

4 By using the substitution $u = e^x$, solve the equation $2e^{2x} - 5e^x + 2 = 0$, leaving solutions in exact form. [6]

5 A polynomial $f(x) = 4x^2 + ax^2 + bx - 5$ where a and b are constants. When $f(x)$ is divided by $(x - 2)$ it leaves a remainder of 7 and when divided by $(x + 1)$ it leaves a remainder of -5. Find a and b . [7]

6 Express $\frac{3x^3 + 18x^2 + 19x + 20}{(x+3)(x^2+2)}$ in partial fractions. [7]

7 The table below shows the experimental values of two variables x and y .

x	1	4	10	25	40
y	1.6	2	2.6	3.8	4.7

Given that x and y are related by the equation $y^2 = ax - b$ where a and b are constants,

(a) draw a straight line graph of y^2 against x to represent the data given in the table above. [4]

(b) use the graph to estimate the value of a and b correct to 1 decimal place. [3]

8 The function $f(x)$ is denoted by

$$f(x) = 5 - \ln(x - 4), x \in \mathbb{R}, x > 4$$

(a) Sketch the graph of $f(x)$ showing clearly where it crosses the axis. [3]

(b) Hence or otherwise state the range of $f(x)$. [1]

(c) Find the inverse of $f(x)$ stating clearly the domain. [3]

$$y = \ln$$

28/5

26
-12

- 9 (a) Given that

$$\sum_{r=1}^{30} (5a + 4r) = 3960$$

[4]

where a is a constant. Find the value of a .

- (b) Using the value of a found in (a), find an expression in terms of n for

$$\sum_{r=5}^n (5a + 4r)$$

[2]

- 10 (a) (i) Sketch the graphs of $y = \sin x$ and $y = \frac{\pi}{2} - 2x$ for $0 \leq x \leq \frac{\pi}{2}$ on the same axis. [3]

- (ii) Determine the number of real roots of the equation $2x - \frac{\pi}{2} + \sin x = 0$ [1]

- (b) Show that there is a root of the equation $2x - \frac{\pi}{2} + \sin x = 0$ between 0.51 and 1.6. [3]

- 11 Given that $y = e^{3x} \sin 5x$

- (a) Find

Diff y for $\cos$

(i) $\frac{dy}{dx}$ [2]

(ii) $\frac{d^2y}{dx^2}$ [2]

- (b) Hence show that

$$\frac{d^2y}{dx^2} + 3 \frac{dy}{dx} - 20y = 9e^{3x} [5 \cos 5x - 3 \sin 5x].$$

[3]

- 12 By use of the substitution $u = x^3 + 4$ evaluate

$$\int_2^{31} \frac{5x^2}{(x^3 + 4)^2} dx$$

giving the answer correct to three decimal places.

[8]

ln x

Sub $u = x^3 + 4$

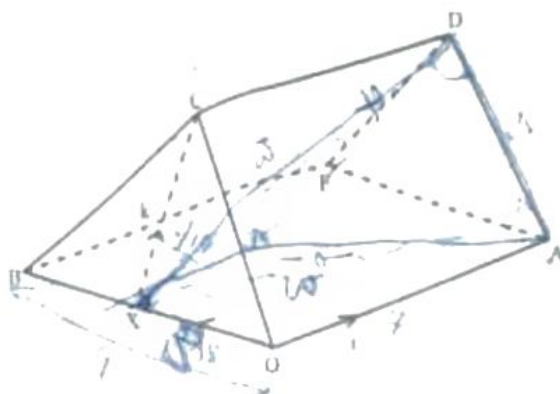
Integrate

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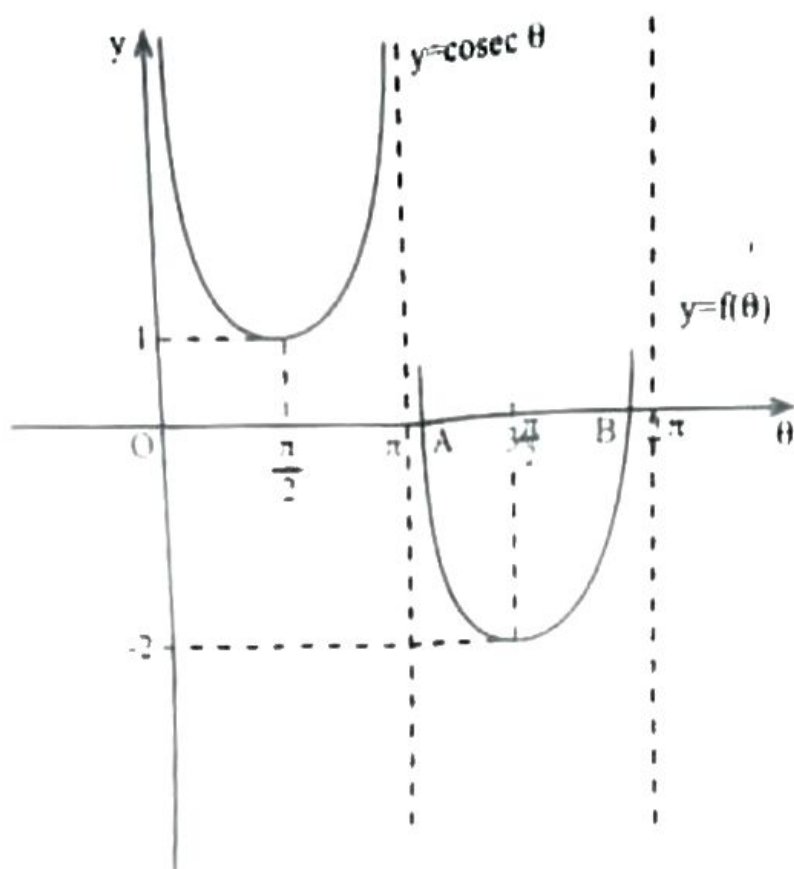
13

The diagram shows a rectangular prism with a fixed origin O . The unit vectors i, j and k are parallel to OA, OB and OC respectively. The point X is such that $OX = \frac{1}{3}OB$ and CX is perpendicular to OB . The lengths of OA, OB and OC are 7 units, 9 units and 5 units respectively



Find

- (a) $\overline{XC}$, [3]
- (b) the cosine of $\angle XDA$, [3]
- (c) the exact area of triangle XDA . [4]
- 14 (a) Given that $z + 2$ is a factor of $f(z) = z^3 - 2z^2 - z + 14$, factorise $f(z)$ completely. [3]
- (b) (i) Solve the equation $f(z) = 0$. [4]
- (ii) Hence or otherwise show the roots on an argand diagram. [3]
- 15 During stages of cell growth in a laboratory, the rate of number of cells x present at any particular time t minutes increases proportionally to $(20 - x)$.
- (a) Form a differential equation. [2]
- (b) If initially the laboratory contains 10 cell growths and after 1 hr 30 minutes the cell growth increases to 18 calculate the number of cells after 2 hours giving the answer to 2 significant figures. [9]



The diagram above shows two graphs of $y = \text{Cosec}\theta$ for $0 \leq \theta \leq \pi$ and $y = f(\theta)$ for $\pi \leq \theta \leq 2\pi$. The graph of $y = f(\theta)$ is as a result of the graph of $y = \text{Cosec}\theta$ by two single transformations

- Write down the resulting equation represented by the diagram describing clearly the transformations done [3]
- Using the result of the equation found in (a) find the values of A and B in terms of π [8]

$$\frac{z+2}{z^3}$$

$$\frac{d}{dx} \left(\frac{1}{x^3} \right) = -\frac{3}{x^4}$$

DE MANUE SOLUTIONS

SUGGESTED MARKING GUIDE

ZIMSEC NOVEMBER 2023 PAPER 1

PURE MATHEMATICS

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1. Since the curve $y = 16x - 17$ is a tangent to the curve $y = 3x^2 + 4x - 5$, the discriminant is $b^2 - 4ac = 0$, implying that there are two repeated roots

Now ... equating the values of y

$$3x^2 + 4x - 5 = 16x - 17$$

$$3x^2 + 4x - 16x - 5 + 17 = 0$$

$$3x^2 - 12x + 12 = 0$$

$$\text{Since } b^2 - 4ac = 0$$

$$a = 3; b = -12 \text{ and } c = 12$$

$$\text{Then } (-12)^2 - 4(3)(12) = 0$$

$$144 - 144 = 0$$

$$0 = 0 \text{ As required}$$

$$2. 4\sin\left(x + \frac{\pi}{6}\right) - 8\cos\left(x - \frac{\pi}{6}\right)$$

Now applying the double angle identities to expand $\sin\theta$ and $\cos\theta$

$$4\left[\sin x \cos \frac{\pi}{6} + \cos x \sin \frac{\pi}{6}\right] - 8\left[\cos x \cos \frac{\pi}{6} + \sin x \sin \frac{\pi}{6}\right]$$

$$= 4\left(\frac{\sqrt{3}}{2}\sin x + \frac{1}{2}\cos x\right) - 8\left(\frac{\sqrt{3}}{2}\cos x + \frac{1}{2}\sin x\right)$$

$$= 2\sqrt{3}\sin x + 2\cos x - 4\sqrt{3}\cos x - 4\sin x$$

When x is small: $\sin x \approx x$ and $\cos x \approx 1 - \frac{x^2}{2}$

$$= 2\sqrt{3}x + 2\left(1 - \frac{x^2}{2}\right) - 4\sqrt{3}\left(1 - \frac{x^2}{2}\right) - 4x$$

$$= 2\sqrt{3}x + 2 - x^2 - 4\sqrt{3} + 2\sqrt{3}x^2 - 4x$$

$$= (2\sqrt{3} - 1)x^2 + (2\sqrt{3} - 4)x + 2 - 4\sqrt{3} \text{ shown}$$

3. $x = 2\sin\theta + 4$

$$\frac{x - 4}{2} = \sin\theta$$

$$\sin^2\theta = \left(\frac{x - 4}{2}\right)^2$$

$$y = \cos 2\theta$$

$$y = 2\cos^2\theta - 1$$

$$\frac{y + 1}{2} = \cos^2\theta$$

Applying the primary identity: $\cos^2\theta + \sin^2\theta = 1$

$$\frac{y + 1}{2} + \left(\frac{x - 4}{2}\right)^2 = 1$$

$$2y + 2 + x^2 - 8x + 16 = 4$$

$$2y = -x^2 + 8x - 14$$

4. Given that $u = e^x$

Now substituting u into the equation

$$2u^2 - 5u + 2 = 0$$

$$2u^2 - 4u - u + 2 =$$

$$2u(u - 2) - 1(u - 2) = 0$$

$$(2u - 1)(u - 2) = 0$$

$$u = \frac{1}{2} \text{ or } u = 2$$

Since $u = e^x$, then $e^x = 2$ or $e^x = \frac{1}{2}$

$$x = \ln 2 \text{ or } x = \ln\left(\frac{1}{2}\right)$$

5. $f(x) = 4x^3 + ax^2 + bx - 5$

$$f(2) = 4(2)^3 + a(2)^2 + b(2) - 5 = 7$$

$$16 + 4a + 2b - 5 = 7$$

$$4a + 2b = -21 \dots \dots \dots (i)$$

$$f(-1) = 4(-1)^3 + a(-1)^2 + b(-1) - 5 = -5$$

$$-4 + a - b - 5 = -5$$

$$a - b = 4$$

$$a = 4 + b \dots \dots \dots (ii)$$

Solving simultaneously

$$4a + 2b = -20$$

$$a - b = 4$$

$$4(4 + b) + 2b = -20$$

$$16 + 4b + 2b = -20$$

$$6b = -36$$

$$b = -6$$

$$a - (-6) = 4$$

$$a = 4 - 6$$

$$a = -2$$

$$\therefore a = -2 \text{ and } b = -6$$

3

6.

$$\begin{array}{r} x^3 + 3x^2 + 2x + 6 \\ 3x^3 + 18x^2 + 19x + 20 \\ -(3x^3 + 9x^2 + 6x + 18) \\ \hline 9x^2 + 13x + 2 \end{array}$$

$$\frac{3x^3 + 18x^2 + 19x + 2}{x^3 + 3x^2 + 2x + 6} = 3 + \frac{9x^2 + 13x + 2}{x^3 + 3x^2 + 2x + 6}$$

$$= 3 + \frac{9x^2 + 13x + 2}{(x + 3)(x^2 + 2)}$$

$$\frac{A}{(x + 3)} + \frac{Bx + c}{(x^2 + 2)}$$

$$A(x^2 + 2) + (Bx + c)(x + 3) = 9x^2 + 13x + 2$$

$$\text{when } x = -3$$

$$81 - 39 + 2 = 11A \quad 44 = 11A$$

$$\text{when } x = 0$$

$$2 = 2A + 3C$$

$$-6 = 3C$$

$$C = -2$$

$$\text{when } x = 1$$

$$24 = 12 + 4B - 8$$

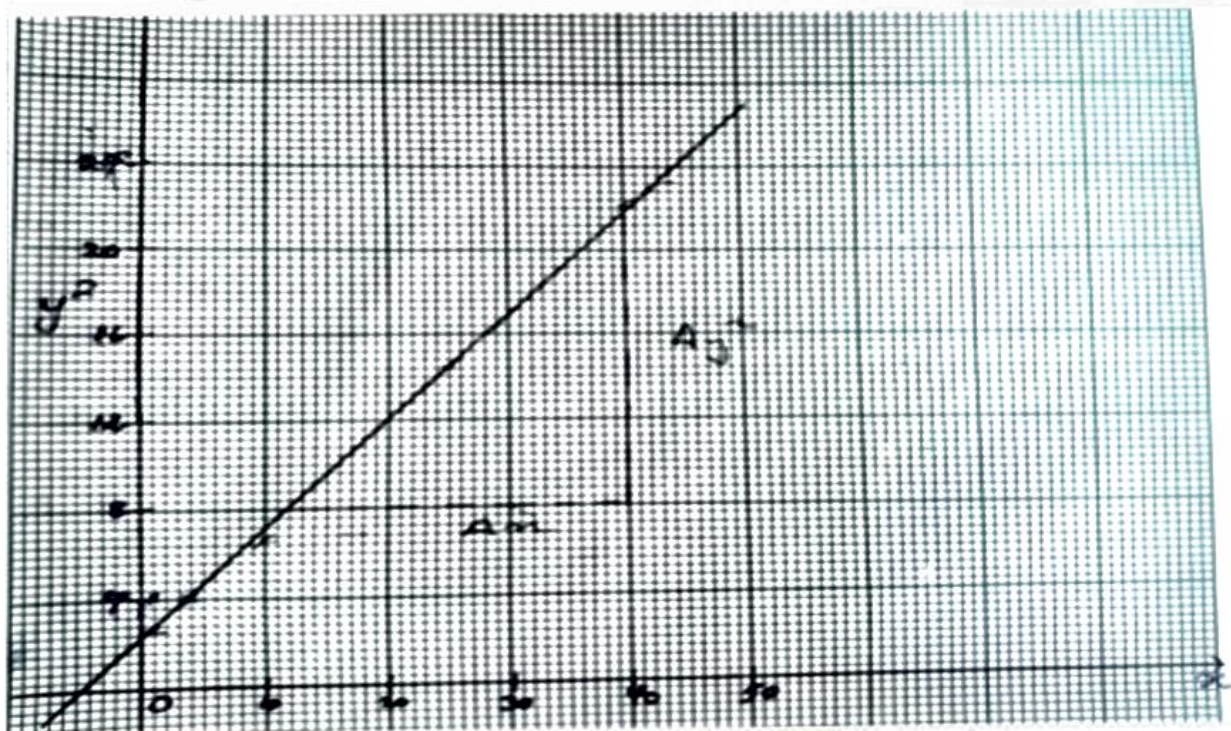
$$20 = 4B$$

$$B = 5$$

$$\frac{3x^3 + 18x^2 + 19x + 2}{x^3 + 3x^2 + 2x + 6} = 3 + \frac{4}{(x + 3)} + \frac{5x - 2}{(x^2 + 2)}$$

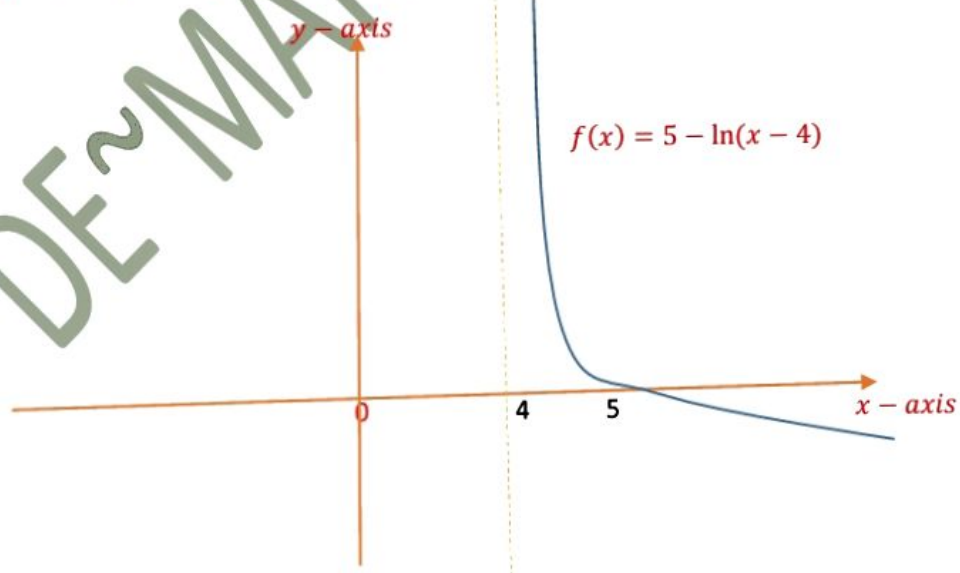
7.

x	1	4	10	25	40
y	1.6	2	2.6	3.8	4.7
y^2	2.56	4	6.76	14.44	22.09



$a = 0.5$ and $b = 2.4$

8. $f(x) = 5 - \ln(x - 4); x \in \mathbb{R}; x > 4$.



b) The range is $f(x) \in \mathbb{R}$

$$c) y = 5 - \ln(x - 4)$$

$$\ln(x - 4) = 5 - y$$

$$e^{\ln(x-4)} = e^{5-y}$$

$$x - 4 = e^{5-y}$$

$$f^{-1}(x) = 4 + e^{5-x}; x \in R$$

9 a)

$$\sum_{r=1}^{30} (5a + 4r) = 3960$$

Checking whether the sequence follows arithmetic progression or geometric progression

$$\text{when } r = 1; 5a + 4$$

$$r = 2; 5a + 8$$

$$r = 3; 5a + 12$$

$$r = 4; 5a + 16$$

There is a common difference of 4 hence the sequence follows AP

$$a = 5a + 4$$

$$d = 4$$

Sum of AP

$$S_n = \frac{n}{2} (2a + (n - 1)d)$$

$$S_{30} = 3960$$

$$\frac{30}{2} [2(5a + 4) + 29(4)] = 3960$$

$$5a + 4 + 58 = 132$$

$$5a = 132 - 62$$

$$5a = 70$$

$$a = 14$$

b)

$$\sum_{r=5}^n (70 + 4r) = \sum_{r=1}^n (70 + 4r) - \sum_{r=1}^4 (70 + 4r)$$

$$\sum_{r=1}^n (70 + 4r) = 70n + 4 \left[\frac{n}{2} (n + 1) \right]$$

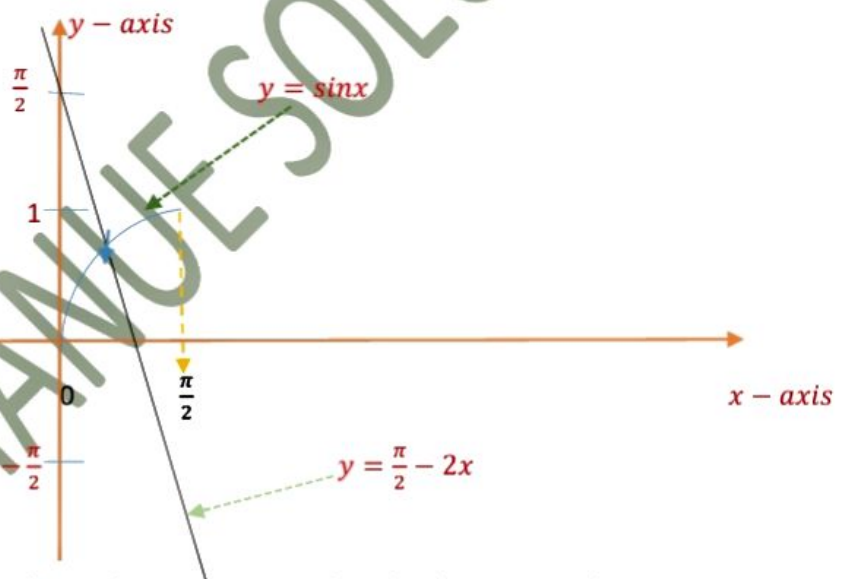
$$= 70n + 2n(n + 1)$$

$$\sum_{r=1}^4 (70 + 4r) = (70 \times 4) + 4 \left[\frac{4}{2} (4 + 1) \right] = 280 + 40 = 320$$

$$\sum_{r=1}^n (70 + 4r) - \sum_{r=1}^4 (70 + 4r) = 70n + 2n(n + 1) - 320$$

$$= 70n + 2n(n + 1) - 320$$

10) a i)



ii) From the graph above, since the graphs intersect one, then they have one real root

b) Let $f(x) = 2x - \frac{\pi}{2} + \sin x$

$$\text{Now; } f(0.51) = 2(0.51) - \frac{\pi}{2} + \sin(0.51) = -0.0626$$

$$f(1.6) = 2(1.6) - \frac{\pi}{2} + \sin(1.6) = 2.6288$$

Since there is a sign change, the root lies between 0.51 and 1.6

11. a i) $y = e^{3x} \sin 5x$

let $u = e^{3x}$ and $v = \sin 5x$

$$\frac{du}{dx} = 3e^{3x}$$

$$\frac{dv}{dx} = 5\cos 5x$$

Now applying the product rule

$$\frac{dy}{dx} = v \frac{du}{dx} + u \frac{dv}{dx}$$

$$\frac{dy}{dx} = \sin 5x(3e^{3x}) + e^{3x}(5\cos 5x)$$

$$= 5e^x \cos 5x + 3e^{3x} \sin 5x$$

ii)

$$\begin{aligned} \frac{d^2y}{dx^2} &= 5e^{3x}(-5\sin 5x) + 15e^{3x}(\cos 5x) + 3[5e^{3x} + 3e^{3x}\sin 5x] \\ &= -25e^{3x}\sin 5x + 15e^{3x}\cos 5x + 15e^{3x}\cos 5x + 9e^{3x}\sin 5x \\ &= 30e^{3x}\cos 5x - 16e^{3x}\sin 5x \end{aligned}$$

$$b) \frac{d^2y}{dx^2} + 3\frac{dy}{dx} - 20y = 9e^{3x}(5\cos 5x - 3\sin 5x)$$

$$\begin{aligned} LHS &= 30e^{3x}\cos 5x - 16e^{3x}\sin 5x + 3[5e^x \cos 5x + 3e^{3x}\sin 5x] - 20e^{3x}\sin 5x \\ &= 45e^{3x}\cos 5x - 27e^{3x}\sin 5x \end{aligned}$$

$$9e^{3x}(5\cos 5x - 3\sin 5x) = RHS; \text{ shown}$$

12.

$$\int_{12}^{31} \frac{5x^2}{(x^3 + 4)^{\frac{1}{2}}} dx$$

$$u = x^3 + 4$$

$$\frac{du}{dx} = 3x^2$$

$$dx = \frac{du}{3x^2}$$

change of limits

$$u = (31)^3 + 4 = 29795$$

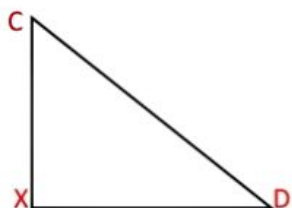
$$u = (12)^3 + 4 = 1732$$

$$\int_{1732}^{29795} \left(\frac{5x^2 \times \frac{du}{3x^2}}{u^{\frac{1}{2}}} \right)$$

$$= \frac{5}{3} \int_{1732}^{29795} u^{-\frac{1}{2}}$$

$$\begin{aligned}
 &= \frac{5}{3} \left[\frac{u^{-\frac{1}{2}+1}}{\frac{1}{2}} \right] = \frac{10}{3} \sqrt{u} \\
 &= \frac{10}{3} (\sqrt{29795} - \sqrt{1732}) \\
 &= 436.650 \text{ (3dp)}
 \end{aligned}$$

13. a) Considering triangle CDX



$$XC = \sqrt{5^2 - 3^2}$$

$$XC = \sqrt{16} = 4$$

$$\vec{XC} = 4\mathbf{i}$$

$$\vec{XD} = \vec{CD} + \vec{XC}$$

$$= 7\mathbf{i} + 4\mathbf{k}$$

$$\vec{AD} = 3\mathbf{j} + 4\mathbf{k}$$

$$\cos \widehat{XDA} = \frac{16}{5\sqrt{65}}$$

$$\text{c) Area} = \frac{1}{2} \|\vec{XD} \times \vec{AD}\| = \frac{1}{2} \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 7 & 0 & 4 \\ 0 & 3 & 4 \end{vmatrix}$$

$$= \frac{1}{2} (\sqrt{(-12)^2 + (-28)^2 + (21)^2})$$

$$= \frac{1}{2} (37)$$

$$= \frac{37}{2} \text{ square units}$$

Accept any other method

14) a) $f(z) = z^3 - 2z^2 - z + 14$

$$\begin{array}{r}
 z^2 - 4z + z \\
 \underline{z + 2} \quad z^3 - 2z^2 - z + 14 \\
 -(z^3 + 2z^2) \\
 \hline
 -4z^2 - z \\
 -(-4^2 - 8z) \\
 \hline
 7z + 14 \\
 -(7z + 14) \\
 \hline
 0 \quad 0
 \end{array}$$

$$f(z) = (z + 2)(z^2 - 4z + 7)$$

b i) Applying the quadratic formula to solve the equation

$$z = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$a = 1; b = -4; c = 7$$

$$= \frac{-(-4) \pm \sqrt{(-4)^2 - 4(1 \times 7)}}{2(1)}$$

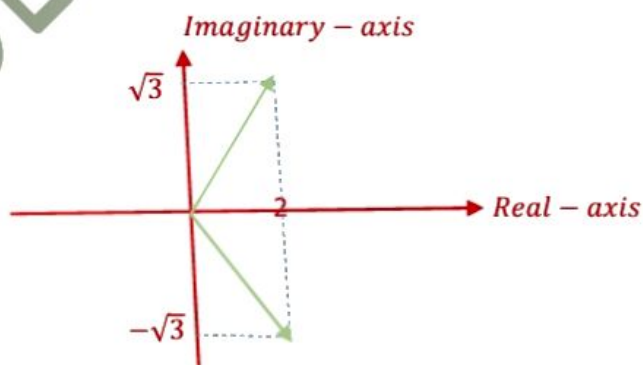
$$= \frac{4 \pm \sqrt{16 - 28}}{2}$$

$$= \frac{4 \pm 2\sqrt{3}i}{2}$$

$$= 2 \pm \sqrt{3}i$$

$\therefore$ The roots of $f(z)$ are : $z = -2; 2 + \sqrt{3}i; 2 - \sqrt{3}i$

ii)



15 a)

$$\frac{dx}{dt} \propto (20 - x)$$

$$\frac{dx}{dt} = k(20 - x)$$

b) Initially; $t = 0, x = 10$

90 minutes after; $x = 18$

$$\int \frac{1}{20 - x} dx = \int k dt$$

$$-\ln(20 - x) = kt + c$$

$$\ln(20 - x) = -kt - c$$

$$e^{\ln(20 - x)} = e^{(-kt - c)}$$

$$20 - x = e^{-kt} \cdot e^{-c}$$

$$x = 20 - Ae^{kt} \text{ where } A = e^{-c}$$

applying the initial condition

$$t = 0 \text{ when } x = 10$$

$$20 - 10 = 20 - Ae^{-k(0)}$$

$$10 = 20 - A$$

$$A = 10$$

$$x = 20 - 10e^{-kt}$$

After 90 minutes

$$t = 90 \text{ when } x = 18$$

$$18 = 20 - 10e^{-90k}$$

$$10e^{-90k} = 2$$

$$e^{-90k} = 0.2$$

$$-90k = \ln(0.2)$$

$$k = \frac{-1}{90} \ln(0.2)$$

$$k = -\ln(0.2)^{\frac{1}{90}}$$

$$x = 20 - 10e^{\ln(0.2)^{\frac{t}{90}}}$$

when $t = 1 \text{ hour and half (120 minutes)}$

$$x = 20 - 10e^{\ln(0.2)^{\frac{120}{90}}}$$

$$x = 19$$

16 a) -Translation along the θ - axis(x - axis) with the translation vector $\begin{pmatrix} \pi \\ 0 \end{pmatrix}$

- Translation along the y - axis with the translation vector $\begin{pmatrix} 0 \\ -3 \end{pmatrix}$

Therefore the resulting equation

$$= \operatorname{cosec}(\theta - \pi) - 3$$

b)

$$\operatorname{cosec}(\theta - \pi) - 3 = 0$$

$$\frac{1}{\sin(\theta - \pi)} = 3$$

$$\frac{1}{3} = \sin(\theta - \pi)$$

$$\theta - \pi = \sin^{-1}\left(\frac{1}{3}\right)$$

$$\therefore A = \pi - \sin^{-1}\left(\frac{1}{3}\right) \text{ and } B = 2\pi - \sin^{-1}\left(\frac{1}{3}\right)$$

Proverbs 16 vs 3 – E.Zivanayi. (De~Manue Solutions)

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FREE TO CONTACT ME SO THAT WE
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ZIMBABWE SCHOOL EXAMINATIONS COUNCIL

General Certificate of Education Advanced Level

PURE MATHEMATICS
PAPER 2

6042/2

NOVEMBER 2023 SESSION

3 hours

Additional materials:

Answer paper

Graph paper

List of Formulae MF 7

Scientific calculator [Non-programmable]

TIME 3 hours

INSTRUCTIONS TO CANDIDATES

Write your Name, Centre number and Candidate number in the spaces provided on the answer paper/answer booklet.

Answer **all** questions in Section A and any **five** questions from Section B.

If a numerical answer cannot be given exactly and the accuracy required is not specified in the question, then in the case of an angle it should be given correct to the nearest degree, and in other cases it should be given correct to 2 significant figures.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question or part question.

The total number of marks for this paper is 120.

The use of a non-programmable scientific calculator is expected, where appropriate.

You are reminded of the need for clear presentation in your answers.

This question paper consists of 4 printed pages.

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[Turn over

Section A (40 marks).

Answer all questions in this section.

1 (a) Simplify $\frac{3^{2x-3} \cdot 81^{x+\frac{1}{2}}}{27^x}$. [3]

(b) Hence or otherwise solve the equation

$$\frac{3^{2x-3} \cdot 81^{x+\frac{1}{2}}}{27^x} = \frac{1}{5^{x-1}}$$

giving the answer in exact form. [4]

2 (a) Show that 1 is a root of the polynomial $Q(x) = 2x^3 + 3x^2 - 4x - 1$. [2]

(b) Hence find the polynomial $P(x)$ such that $Q(x) = (x - 1)P(x)$ [3]

(c) (i) Express $P(x)$ in the form $A(x + B)^2 + C$. [3]

(ii) Hence state the minimum value of $P(x)$. [1]

3 The curved surface area, A of a cylindrical drum varies jointly as its circular base diameter d , and its height l . Given that the curved surface area of the drum is 2.64m^2 when its base radius is 0.35m and its height is 1.2m ,

(a) find the relationship between A , d and l , [5]

(b) calculate the radius of a cylinder whose surface area is 14.52m^2 and height is 3m to 2 decimal places. [4]

4 Given that $f(x) = kx^2 + 3x + 3$ and $g(x) = kx + 7$, find the range of values of k for which $f(x) = g(x)$ has two distinct solutions. [5]

5 (a) Given that $2x^3 + ax^2 + x - 3 \equiv (bx^2 + c)(x - 3)$, find the constants a , b and c . [4]

(b) Hence express $\frac{1+6x}{2x^3+ax^2+x-3}$ in partial fractions. [6]

Section B (80 marks)

Answer any five questions from this section. Each question carries 16 marks.

- 6 Functions $f(x)$, $t(x)$ and $h(x)$ are defined by

$$f(x) = 12x + 20$$

$$t(x) = e^x$$

$$h(x) = 2x + 3.$$

- (a) (i) Find $th(x)$. [1]
- (ii) On the same axes, sketch the graphs of $y = f(x)$ and $y = th(x)$. [5]
- (b) Calculate by integration the area between the graph of $y = f(x)$, $y = th(x)$ and the lines $x = -1.5$ and $x = 0$. [3]
- (c) (i) Estimate the area in (b) using the trapezium rule with 4 ordinates. [4]
- (ii) Calculate the relative error made in estimating the area in (b) using the trapezium rule. [3]

- 7 (a) Prove the Identity $\frac{2 \cot 2\theta}{1 - \tan^2 \theta} \equiv \cot \theta$. [4]

- (b) Hence or otherwise solve the equation $\frac{2 \cot 2\theta}{1 - \tan^2 \theta} = 2 \operatorname{cosec}^2 \theta - 1$, for θ in the range $0^\circ \leq \theta \leq 360^\circ$. [12]

- 8 Matrices A and B^{-1} are given by

$$A = \begin{pmatrix} 1 & 2 & 1 \\ x & 1 & x+1 \\ 2 & x-2 & 0 \end{pmatrix}, x > 0$$

$$B^{-1} = \frac{1}{8} \begin{pmatrix} 1 & -1 & 1 \\ 2 & 1 & 0 \\ 1 & 0 & 3 \end{pmatrix}$$

- (a) Calculate the value of x given that the determinant of A is equal to 13 [5]

- (b) Hence find

(i) A^{-1} . [7]

(ii) $(AB)^{-1}$. [4]

- 9 The sum of n terms of a sequence, S_n is given by

$$S_n = \frac{n}{4}(5 - n), \text{ for } n = 1; 2; 3 \dots$$

- (a) Find the second and third terms of the sequence. [4]
- (b) Hence or otherwise, find the general term of the sequence in term of n . [3]
- (c) Calculate the number of terms needed for the Sum, S_n , to be equal to -9 . [5]
- (d) Find the range of value of n for which $S_n > 0$. [4]

- 10 A quadrilateral ABCD has vertices at points A(-1;1); B(3;7); C(4;2) and D(2; -1), respectively.

- (a) Find the equation of line AB. [4]
- (b) Show that line AB is perpendicular to line AD. [4]
- (c) Find the equation of a circle with center at point D and radius CD in the form $x^2 + y^2 + ax + by = c$ where a , b and c are constants. [5]
- (d) Show that the circle in (c) passes through point A. [2]
- (e) Explain why line AB is a tangent to the circle in (c). [1]

- 11 Triangle ABC has vertices with position vectors $\mathbf{i} + 2\mathbf{j} + 2\mathbf{k}$, $2\mathbf{i} + 5\mathbf{j} - 2\mathbf{k}$ and $5\mathbf{i} + 4\mathbf{j} + \mathbf{k}$ respectively.

Find

- (a) $\overrightarrow{AC}$, [2]
- (b) the cartesian equation of line AC, [4]
- (c) (I) the perpendicular distance of point B from line AC, [8]
- (II) the area of triangle ABC. [2]

- 12 (a) Express $g(x) = \frac{4x+5}{2(x+1)}$ in the form $A + \frac{B}{2(x+1)}$, where A and B are constants. [2]
- (b) With the aid of diagrams, in each case, illustrate the transformations which maps the graph of $y = \frac{1}{x}$ on to the graph of $y = g(x)$. [10]
- (c) Find the inverse of $g(x)$ and state its domain. [4]

DE MANUE SOLUTIONS

SUGGESTED MARKING GUIDE

ZIMSEC NOVEMBER 2023 PAPER2

PURE MATHEMATICS

6042/2

0717256148 / 0774459409

1.a) $\frac{3^{2x-3} \cdot 81^{x+\frac{1}{2}}}{27^x}$

$$= 3^{2x-3} \cdot 4^{4\left(x+\frac{1}{2}\right)} \cdot 3^{-3x}$$

Aside simplifying the powers

$$2x - 3x + 4\left(x + \frac{1}{2}\right) - 3x$$

$$2x - 3x + 4x + 2 - 3x$$

$$\text{Now : } 3^{3x-1}$$

$$3^{3x} \cdot 3^{-1}$$

$$= \frac{27^x}{3}$$

b)

$$\frac{3^{2x-3} \cdot 81^{x+\frac{1}{2}}}{27^x} = \frac{1}{5^{x-1}}$$

$$\frac{27^x}{3} = 5^{-(x-1)}$$

$$(3x - 1)\ln 3 = (1 - x)\ln 5$$

$$3x\ln 3 - \ln 3 = \ln 5 - x\ln 5$$

$$x(3\ln 3 + \ln 3) = \ln(5 \times 3)$$

$$x \ln(135) = \ln 15$$

$$x = \frac{\ln 15}{\ln 135}$$

$$2a) G(x) = 2x^3 + 3x^2 - 4x - 1$$

If 1 is a root then $G(1)$ should give us zero

$$G(1) = 2(1)^3 + 3(1)^2 - 4(1) - 1$$

$$= 2 + 3 - 4 - 1$$

$$= 0 \text{ as required}$$

b) Since $G(1)$ is 0 then $x - 1$ is a factor

$$\begin{array}{r}
 2x^2 + 5x + 1 \\
 x - 1 \overline{) 2x^3 + 3x^2 - 4x - 1} \\
 \underline{-(2x^3 - 2x^2)} \\
 5x^2 - 4x \\
 \underline{-(5x^2 - 5x)} \\
 x - 1 \\
 \underline{-(x - 1)} \\
 0
 \end{array}$$

$$G(x) = (x - 1)(2x^2 + 5x + 1)$$

$$\therefore P(x) = (2x^2 + 5x + 1)$$

$$C i) P(x) = 2x^2 + 5x + 1$$

$$= 2 \left[x^2 + \frac{5}{2}x + \frac{1}{2} \right]$$

$$= 2 \left[\left(x + \frac{5}{4} \right)^2 + \frac{1}{2} - \frac{25}{16} \right]$$

$$= 2 \left(x + \frac{5}{4} \right)^2 - \frac{17}{8}$$

C ii) Minimum value

$$= -\frac{17}{8}$$

3 .a)

$$A \propto dl$$

$$A = kdl$$

$$2.64 = k(2 \times 0.35)(1.2)$$

$$k = \frac{2.64}{(2 \times 0.35 \times 1.2)}$$

$$k = \frac{22}{7}$$

$$A = \frac{22}{7} dl$$

b) $r = \frac{d}{2}$

$$d = \frac{7A}{22l}$$

$$d = \frac{7(14,52)}{22(3)}$$

$$d = 1.54$$

$$r = 1.54 \div 2$$

$$r = 0,77$$

$$r = 0,77m$$

4) $f(x) = kx^2 + 3x + 3$

$$G(x) = kx + 7$$

$$kx^2 + 3x + 3 = kx + 7$$

$$kx^2 + 3x - kx - 4 = 0$$

When $f(x) = g(x)$ has two distinct roots then

$$b^2 - 4ac > 0$$

$$(3 - k)^2 - 4(k)(-4) > 0$$

$$= 9 - 6k + k^2 + 16k > 0$$

$$= k^2 + 10k + 9 > 0$$

$$k^2 + k + 9k + 9 > 0$$

$$k(k+1) + 9(k+1) > 0$$

$$(k+9)(k+1) > 0$$

Critical values : $k = -9 ; k = -1$



Solution Set : $k < -9 \cup k > -1$

5 a) $2x^3 + ax^2 + x - 3 \equiv (bx^2 + c)(x - 3)$

$$2x^3 + ax^2 + x - 3 = bx^3 - 3bx^2 + Cx - 3c \quad \text{Comparing Coeffs}$$

$$2 = b ; a = -3(2) = -6 ; c = 1$$

$$a = -6 ; b = 2 \quad \text{and} \quad c = 1$$

b)

$$f(x) = \frac{1 + 6x}{2x^3 + 6x^2 + x - 3}$$

$$\begin{array}{r} 2x^2 + 1 \\ \overline{) 2x^3 + 6x^2 + x - 3} \\ \underline{-(2x^3 - 6x^2)} \\ 12x^2 + x - 3 \\ \underline{-(12x^2 - 6x)} \\ 7x - 3 \\ \underline{-(7x - 21)} \\ 18 \end{array}$$

$$f(x) = \frac{1 + 6x}{(x - 3)(2x^2 + 1)}$$

$$\frac{1 + 6x}{(x - 3)(2x^2 + 1)} \equiv \frac{A}{(x - 3)} + \frac{B}{(2x^2 + 1)}$$

$$1 + 6x = A(2x^2 + 1) + (Bx + c)(x - 3)$$

when $x = 3$

$$19 = 19A$$

$$A = 1$$

when $x = 0 : 1 - 2C$

$$C = 0 \quad x = 1$$

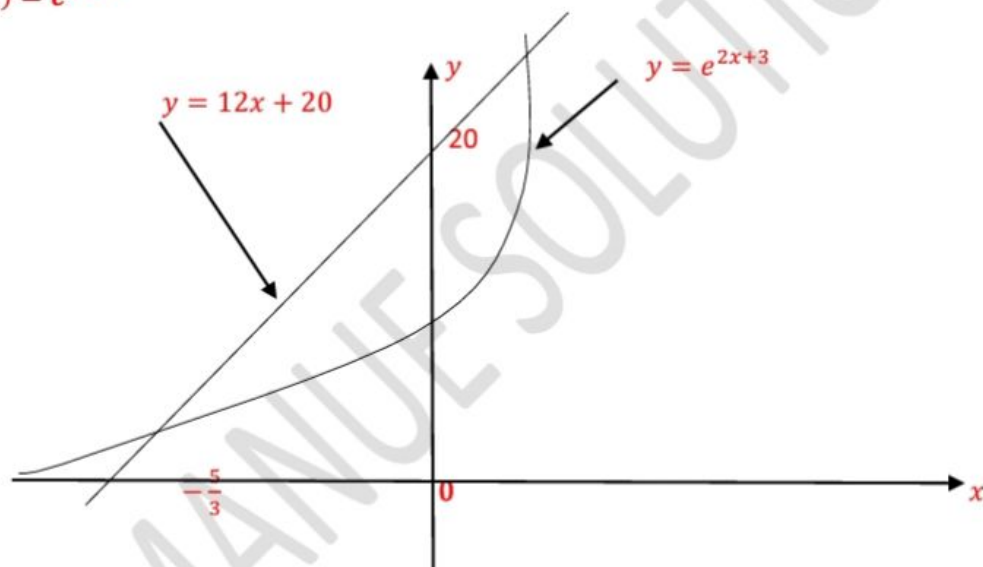
$$7 = 3 - 2B + 0$$

$$B = -2$$

$$f(x) = \frac{1}{x-1} - \frac{2x}{2x^2+1}$$

$$6ai) \text{th}(x) = e^{2x+3}$$

ii)



$$b) \int_{-1.5}^0 ((12x + 20) - (e^{2x+3})) dx$$

$$\left(6x^2 + 20x - \frac{e^{2x+3}}{2} \right) \Big|_{-1.5}^0$$

$$\left(-\frac{e^3}{2} \right) - \left(6(-1.5) + 20(-1.5) - \frac{e^0}{2} \right)$$

$$17 - \frac{e^3}{2} \text{ units}^2$$

$$ci) h = \frac{\text{upp} - \text{low}}{\text{number of trapezia}}$$

$$h = \frac{0 + 1.5}{3}$$

$$h = \frac{1}{2}$$

$x_0 = -1.5$	$y_0 = 1$
$x_1 = -1$	$y_1 = 8 - e^1$
$x_2 = -0.5$	$y_2 = 14 - e^2$
$x_3 = 0$	$y_3 = 20 - e^3$

$$A = \frac{h}{2}(y_0 + y_3 + 2(y_1 + y_2))$$

$$A = \frac{1}{4}(1 + 20 - e^3 + 2(8 - e^1 + 14 - e^2))$$

$$A = 6.17496806$$

$$\mathbf{6.2 \text{ units}^2}$$

$$\text{ii) Relative Error} = \left| \frac{\left(17 - \frac{e^3}{2}\right) - 6.17}{\left(17 - \frac{e^3}{2}\right)} \right|$$

$$= \mathbf{0.11}$$

$$7a) \frac{2\cot 2\theta}{1 - \tan^2 \theta} = \cot \theta$$

Proving on LHS

$$\frac{\frac{2\cos 2\theta}{\sin 2\theta}}{1 - \frac{\sin^2 \theta}{\cos^2 \theta}}$$

$$\frac{\frac{2\cos 2\theta}{\sin 2\theta}}{\frac{\cos^2 \theta - \sin^2 \theta}{\cos^2 \theta}}$$

$$\frac{2\cos 2\theta}{2\sin \theta \cos \theta} \times \frac{\cos^2 \theta}{\cos 2\theta}$$

$$\frac{\cos \theta}{\sin \theta} = \cot \theta$$

$$LHS = RHS \quad \text{shown}$$

$$b) \cot \theta = 2 \operatorname{cosec}^2 \theta - 1$$

$$\cot \theta = 2(\cot^2 \theta + 1) - 1$$

$$\cot \theta = 2\cot^2 \theta + 1$$

$$\text{Let } x = \cot \theta$$

$$2x^2 - x + 1 = 0$$

$$x = \frac{1 \pm \sqrt{(-1)^2 - 4(2)}}{2(2)}$$

The solution has no real roots.

8a) Using the basket weave formula

$$\begin{pmatrix} 1 & 2 & 1 \\ x & 1 & x+1 \\ 2 & x-2 & 0 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ x & 1 \\ 2 & x-2 \end{pmatrix}$$

$$13 = (4(x+1) + x(x-2)) - (2 + (x-2)(x+1))$$

$$13 = (4x + 4 + x^2 - 2x - 2 - (x^2 - x - 2))$$

$$13 = 3x + 4$$

$$9 = 3x$$

$$x = 3$$

$$b) A = \begin{pmatrix} 1 & 2 & 1 \\ 3 & 1 & 4 \\ 2 & 1 & 0 \end{pmatrix}$$

Transposing A

$$A^T = \begin{bmatrix} 1 & 3 & 2 & 1 & 3 \\ 2 & 1 & 1 & 2 & 1 \\ 1 & 4 & 0 & 1 & 4 \\ 1 & 3 & 2 & 1 & 3 \\ 2 & 1 & 1 & 2 & 1 \end{bmatrix}$$

$$\text{Adjoint}(A) = \begin{bmatrix} \begin{vmatrix} 1 & 1 \\ 4 & 0 \end{vmatrix} & \begin{vmatrix} 1 & 2 \\ 0 & 1 \end{vmatrix} & \begin{vmatrix} 2 & 1 \\ 1 & 4 \end{vmatrix} \\ \begin{vmatrix} 4 & 0 \\ 3 & 2 \end{vmatrix} & \begin{vmatrix} 0 & 1 \\ 2 & 1 \end{vmatrix} & \begin{vmatrix} 1 & 4 \\ 1 & 3 \end{vmatrix} \\ \begin{vmatrix} 3 & 2 \\ 3 & 2 \end{vmatrix} & \begin{vmatrix} 2 & 1 \\ 2 & 1 \end{vmatrix} & \begin{vmatrix} 1 & 3 \\ 1 & 3 \end{vmatrix} \\ \begin{vmatrix} 3 & 2 \\ 1 & 1 \end{vmatrix} & \begin{vmatrix} 2 & 1 \\ 1 & 2 \end{vmatrix} & \begin{vmatrix} 1 & 3 \\ 2 & 1 \end{vmatrix} \end{bmatrix}$$

$$\text{Adjoint}(A) = \begin{pmatrix} -4 & 1 & 7 \\ 8 & -2 & -1 \\ 1 & 3 & -5 \end{pmatrix}$$

$$A^{-1} = \frac{1}{13} \begin{pmatrix} -4 & 1 & 7 \\ 8 & -2 & -1 \\ 1 & 3 & -5 \end{pmatrix}$$

$$ii) (AB)^{-1} = A^{-1} \times B^{-1}$$

$$\frac{1}{8} \times \frac{1}{13} \begin{pmatrix} -4 & 1 & 7 \\ 8 & -2 & -1 \\ 1 & 3 & -5 \end{pmatrix} \begin{pmatrix} 1 & -1 & 1 \\ 2 & 1 & 0 \\ 1 & 0 & 3 \end{pmatrix}$$

$$\frac{1}{104} \begin{pmatrix} 5 & 5 & 17 \\ 3 & -10 & 5 \\ 2 & 2 & -14 \end{pmatrix}$$

$$9a) S_n = \frac{n}{4}(5 - n)$$

$$S_1 = \frac{1}{4}(5 - 1)$$

$$T_1 = 1$$

$$S_2 = \frac{2}{4}(5 - 2)$$

$$S_2 = \frac{3}{2}$$

$$S_2 = T_1 + T_2$$

$$T_2 = \frac{3}{2} - 1$$

$$T_2 = \frac{1}{2}$$

$$S_3 = \frac{3}{4}(5 - 3)$$

$$S_3 = \frac{3}{2}$$

$$T_3 = S_3 - T_2 - T_1$$

$$T_3 = \frac{3}{2} - 1 - \frac{1}{2}$$

$$T_3 = 0$$

$$\therefore T_3 = 0 \quad \& \quad T_2 = \frac{1}{2}$$

$$b) T_n = a + (n - 1)d$$

$$T_n = 1 + (n - 1) \times -\frac{1}{2}$$

$$T_n = 1 - \frac{n}{2} + \frac{1}{2}$$

$$T_n = \frac{3 - n}{2}$$

$$c) \frac{n}{4}(5 - n) = -9$$

$$n(5 - n) = -36$$

$$n^2 - 5n - 36 = 0$$

$$(n - 9)(n + 5) = 0$$

$$n = 9 \text{ only}$$

$$d) \frac{n}{4}(5 - n) > 0$$

Critical Values: $n = 0$ or $n = 5$

	$n < 0$	$0 < n < 5$	$n > 5$
$\frac{n}{4}$	-	+	+
$5 - n$	+	+	-
$f(n)$	-	+	-

Solution Set : $0 < n < 5$

$$10a) M_{AB} = \frac{7 - 1}{3 + 1}$$

$$M_{AB} = \frac{3}{2}$$

$$Y - Y_1 = M(X - X_1)$$

$$y - 1 = \frac{3}{2}(x + 1)$$

$$2y - 2 = 3x + 3$$

$$2y = 3x + 5$$

$$b) M_{AD} = \frac{1 + 1}{-1 - 2}$$

$$M_{AD} = \frac{-2}{3}$$

$$M_{AD} \times M_{AB}$$

$$\frac{3}{2} \times -\frac{2}{3}$$

-1

Since the product of the gradients = -1, therefore they are perpendicular

$$c) (x - a)^2 + (y - b)^2 = r^2$$

$$r = \sqrt{(4 - 2)^2 + (2 + 1)^2}$$

$$r = \sqrt{13}$$

$$(x - 2)^2 + (y + 1)^2 = (\sqrt{13})^2$$

$$x^2 - 4x + 4 + y^2 + 2y + 1 = 13$$

$$\mathbf{x^2 + y^2 - 4x + 2y - 8 = 0}$$

d) At A(-1; 1)

$$(-1)^2 + 1^2 - 4(-1) + 2(1) - 8 = 0$$

$$1 + 1 + 4 + 2 - 8 = 0$$

$$0 = 0$$

Since LHS = RHS, therefore the circle passes through point A

e) Because discriminant satisfying both equations = 0 hence the line and circle meet once showing that the line is a tangent

$$11a) \overrightarrow{AC} = OC - OA$$

$$\overrightarrow{AC} = \begin{pmatrix} 5 \\ 4 \\ 1 \end{pmatrix} - \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}$$

$$\overrightarrow{AC} = \begin{pmatrix} 4 \\ 2 \\ -1 \end{pmatrix}$$

$$b) r = a + tb$$

$$r = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix} + t \begin{pmatrix} 4 \\ 2 \\ -1 \end{pmatrix}$$

$$r = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} - \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix} = t \begin{pmatrix} 4 \\ 2 \\ -1 \end{pmatrix}$$

$$t = \frac{x - 1}{4} = \frac{y - 2}{2} = 2 - z$$

ci) Let N be the foot = $\begin{pmatrix} 1+4t \\ 2+2t \\ 2-t \end{pmatrix}$

$$\overrightarrow{BN} = \begin{pmatrix} 1+4t \\ 2+2t \\ 2-t \end{pmatrix} - \begin{pmatrix} 2 \\ 5 \\ -2 \end{pmatrix}$$

$$\overrightarrow{BN} = \begin{pmatrix} -1+4t \\ -3+2t \\ 4-t \end{pmatrix}$$

$$\begin{pmatrix} -1+4t \\ -3+2t \\ 4-t \end{pmatrix} \cdot \begin{pmatrix} 4 \\ 2 \\ -1 \end{pmatrix} = 0$$

$$-4 + 16t - 6 + 4t - 4 + t = 0$$

$$t = \frac{2}{3}$$

$$\begin{pmatrix} 1+4\left(\frac{2}{3}\right) \\ 2+2\left(\frac{2}{3}\right) \\ 2-\left(\frac{2}{3}\right) \end{pmatrix}$$

$$\begin{pmatrix} \frac{11}{3} \\ \frac{10}{3} \\ \frac{4}{3} \end{pmatrix}$$

let d be the Distance from B to line AC

$$d = \sqrt{\left(2 - \frac{11}{3}\right)^2 + \left(5 - \frac{10}{3}\right)^2 + \left(-2 - \frac{4}{3}\right)^2}$$

$$d = \frac{5}{3}\sqrt{6} \text{ units}^2$$

$$ii) A = \frac{1}{2} |\overrightarrow{BA} \times \overrightarrow{BC}|$$

$$\overrightarrow{BA} = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix} - \begin{pmatrix} 2 \\ 5 \\ -2 \end{pmatrix} = \begin{pmatrix} -1 \\ -3 \\ 4 \end{pmatrix}$$

$$\overrightarrow{BC} = \begin{pmatrix} 5 \\ 4 \\ 1 \end{pmatrix} - \begin{pmatrix} 2 \\ 5 \\ -2 \end{pmatrix} = \begin{pmatrix} 3 \\ -1 \\ 3 \end{pmatrix}$$

$$A = \frac{1}{2} \begin{vmatrix} i & j & k \\ -1 & -3 & 4 \\ 3 & -1 & 3 \end{vmatrix}$$

$$A = \frac{1}{2} |-5i + 15j + 10k|$$

$$A = \frac{1}{2} \times \sqrt{(-5)^2 + (15)^2 + (10)^2}$$

$$A = \frac{1}{2} \times \sqrt{350}$$

$$A = \frac{5\sqrt{14}}{2} \text{ units}^2$$

$$12a) \frac{4x+5}{2(x+1)} = A + \frac{B}{2(x+1)}$$

$$4x+5 = 2A(x+1) + B$$

$$\text{For } x = -1$$

$$1 = B$$

$$\text{For } x = 0$$

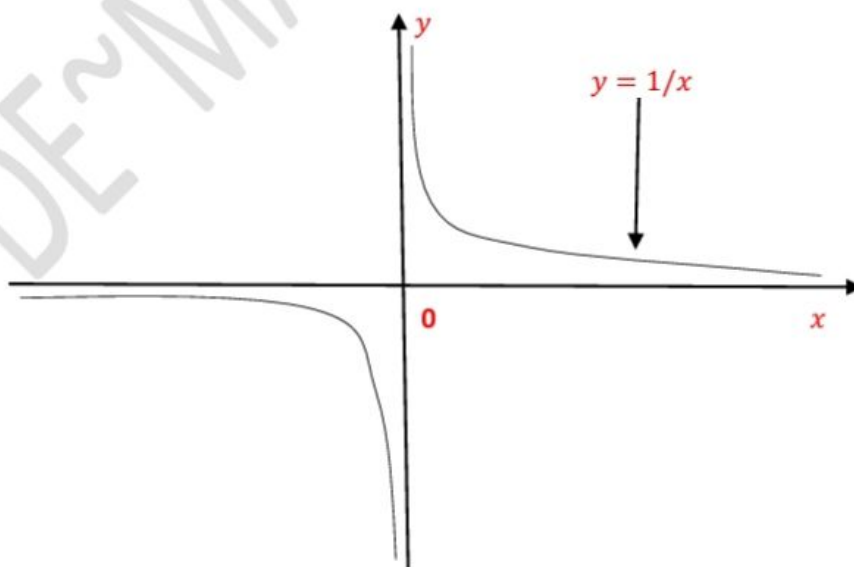
$$5 = 2A + B$$

$$A = \frac{5-1}{2}$$

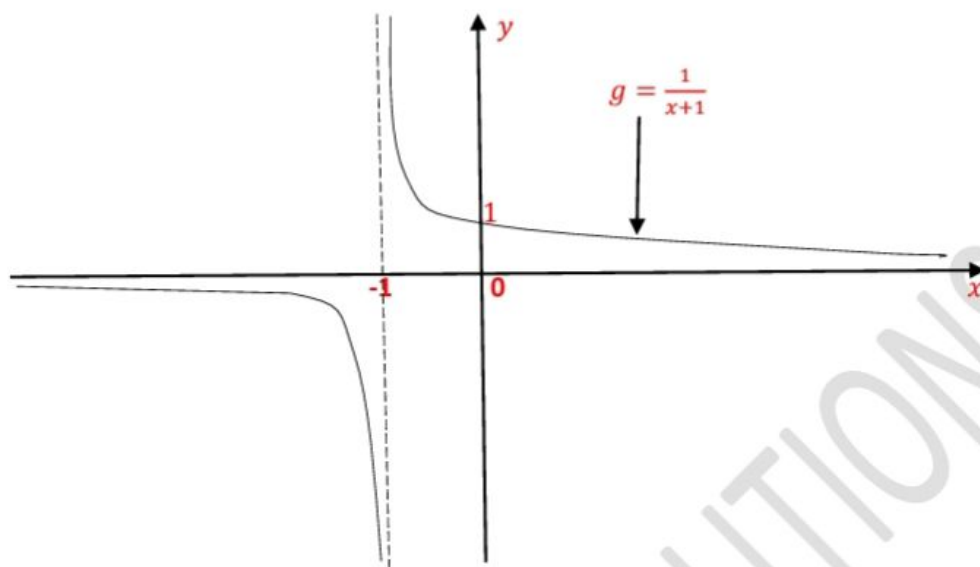
$$A = 2$$

$$2 + \frac{1}{2(x+1)}$$

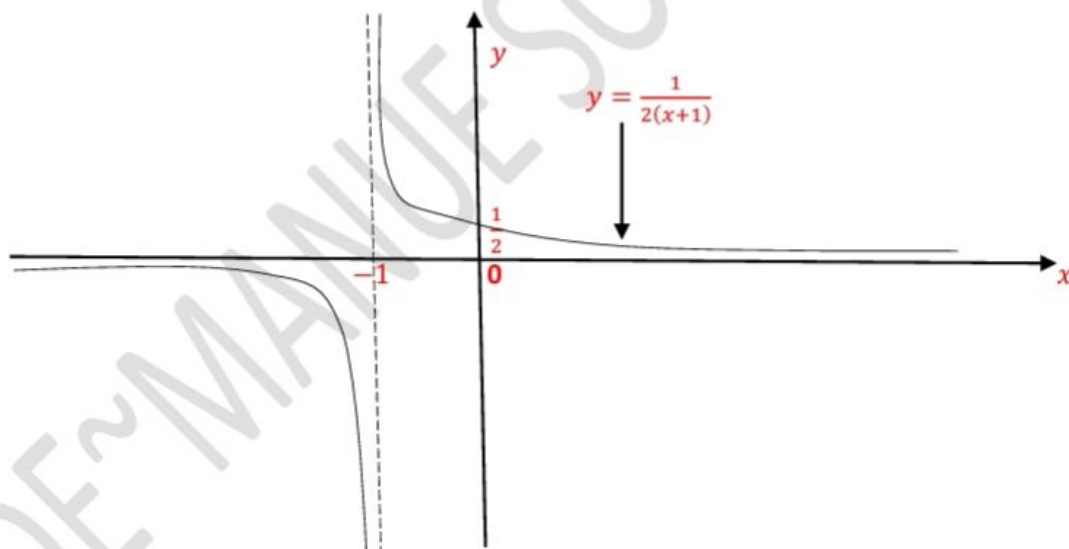
$$b) y = \frac{1}{x}$$



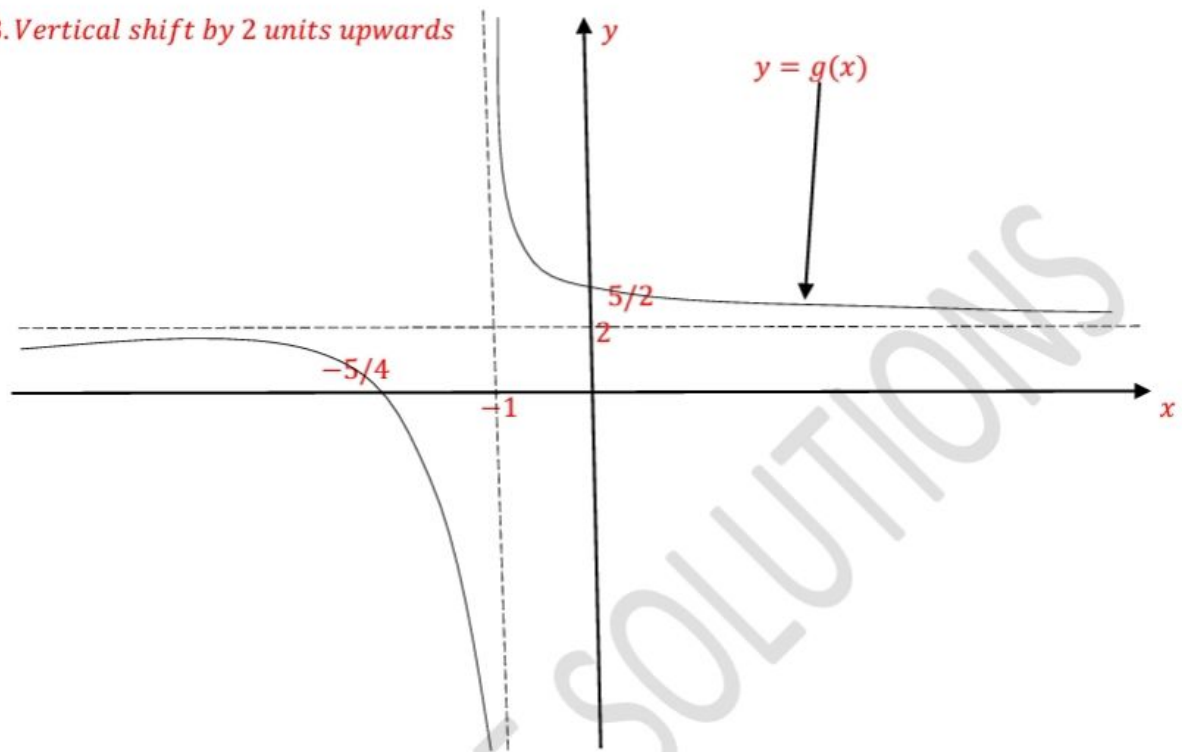
1. Horizontal shift by 1 unit leftwards



2. Stretch of factor $\frac{1}{2}$ parallel to the y – axis



3. Vertical shift by 2 units upwards



$$c) \quad g(x) = \frac{4x + 5}{2(x + 1)}$$

$$y = \frac{4x + 5}{2(x + 1)}$$

$$2y(x + 1) = 4x + 5$$

$$2xy + 2y = 4x + 5$$

$$2y - 5 = 4x - 2xy$$

$$2y - 5 = (4 - 2y)x$$

$$x = \frac{2y - 5}{4 - 2y}$$

$$g^{-1}(x) = \frac{2x - 5}{4 - 2x} \quad x \in \mathbb{R}; x \neq 2$$

CONSTRUCTIVE COMMENTS IN THE FORM OF PRESENTATION, OMMISION AND ERRORS
ARE WELCOME. FEEL FREE TO CONTACT ME SO THAT WE IMPROVE THE DOCUMENT
TOGETHER

CONTACT DETAIL:

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DE ~ MANUE SOLUTIONS

PURE MATHEMATICS

6042/1

PAPER 1

FINAL REVISION OCTOBER 2023 SESSION

3 hours

Additional material

Answer paper

Graph paper

List of Formulae MF7

TIME 3 hours

INSTRUCTION TO CANDIDATES

Write your Name, Centre number and Candidate number in the space provided on the answer paper/booklet.

Answer **all** questions

If a numerical answer cannot be given exactly and accuracy is not specified in question, then in the case of an angle it should be given correct to the nearest degree, and in other cases it should be given correct to 2 significant figures.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question or part question

The total number of marks for this paper is 120

The use of non-programmable calculator is expected, where appropriate

You are reminded of the need for clear presentation in your answer.

This question paper consist of 3 printed pages and one blank

1. Find a polynomial approximation of the function $f(\theta) = \cos^2(\theta)\sin^2(\theta)$ [4]

2 a) Solve the inequality $|x + 3a| > 2|x - 2a|$, where a is a positive constant [4]

b) Express $8 - 3x + 2x^2$ in the form $c + b(x + a)^2$ where a, b and c are constants to be found.

Hence, state the coordinates of the turning points of the graph $8 - 3x + 2x^2$ [4]

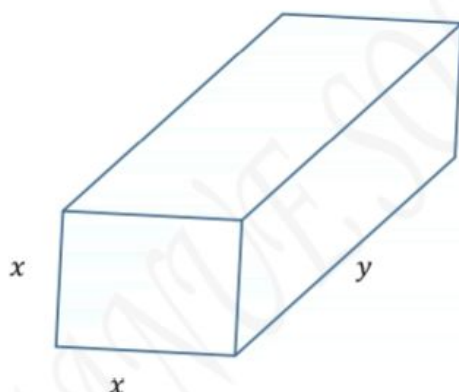
3. Divide $4a^3 - 6a^2b + 5b^3$ by $2a - b$ [5]

4. Express $\frac{\sqrt{1+2x}}{\sqrt[3]{1-3x}}$, as a power series as far as the term in x^2 . Hence state the range of values for which the series is convergent. [6]

5. Solve the equation $3\operatorname{cosec}^2\theta - 5 = 4\cot\theta$ in the range $0 < \theta < 360^\circ$ [5]

6.a) Given that $y = \frac{\sqrt{x}}{\cos x}$, show that $\frac{dy}{dx} = \left[\frac{\frac{\cos x}{2\sqrt{x}} + \sqrt{x}\sin x}{\cos^2 x} \right]$ [6]

b) An open rectangular box with square ends is fitted with an overlapping lid which covers the top and the front face. Determine the maximum volume of the box if 6m^2 of metal are used in construction. [7]



7. Prove by mathematical induction that

$$\sum_{r=1}^n (r^2 + 1)r! = n(n+1)!$$

[6]

8. The variation of resistance R ohms, of an aluminium conductor with temperature $\theta^\circ\text{C}$

is given by $\frac{dR}{d\theta} = \alpha R$, where α is the temperature coefficient of resistance of aluminium.

If $R = R_0$ when $\theta = 0^\circ\text{C}$,

a) Solve the equation for R .

b) If $\alpha = 38 \times 10^{-4}/^\circ\text{C}$, determine the resistance of the aluminium conductor at 50°C , correct to 3 significant figures, when its resistance at 0°C is 24.0 ohms. [11]

9. Relative to the origin O , the position vectors of the points A, B and C are given by

$$\vec{OA} = \begin{pmatrix} 2 \\ 3 \\ -6 \end{pmatrix}, \vec{OB} = \begin{pmatrix} 0 \\ -6 \\ 8 \end{pmatrix} \text{ and } \vec{OC} = \begin{pmatrix} -2 \\ 5 \\ -2 \end{pmatrix}$$

i) Find the angle AOB [4]

ii) Find the vector which is in the same direction as $\vec{AC}$ and has magnitude 30. [3]

iii) Find the value of the constant p for which $\vec{OA} + p\vec{OB}$ perpendicular to $\vec{OC}$. [3]

10. Find the inverse of the matrix $A = \begin{pmatrix} 2 & 1 & 0 \\ 3 & -1 & 4 \\ 1 & 1 & 2 \end{pmatrix}$

Hence solve the system of simultaneous equations

$$2x + y = 1$$

$$3x - y + 4z = 2$$

$$x + y + 2z = -1 \quad [10]$$

11. The complex number $u = \frac{6-3i}{1+2i}$

i) Showing all your workings, find the modulus of u and show that the argument of u is $-\frac{\pi}{2}$ [4]

ii) For complex number z satisfying $\arg(z - u) = \frac{\pi}{4}$, find the least possible value of $|z|$ [3]

iii) For a complex number z satisfying $|z - (1 + i)u| = 1$, find the greatest possible value of $|z|$ [3]

12. a) The square is drawn by joining the mid points of the sides of a given square. A second square is drawn inside the second square in the same way, Second Square is drawn inside the third square and this process continues up to 8^{th} square. If the side of the first square is 32cm then find the area of the first eight squares. [6]

b) The sum of first 10 terms of an arithmetic progression is 120 and the sum of the first 20 terms is 840. Find the sum of first 40 terms of this progress [4]

13. Evaluate

$$\int \frac{1}{(x^2 - a^2)} \quad [5]$$

b) Find the area enclosed by the curve $y = \sin 2x$, the x - axis and the ordinates $x = 0$ and $x = \frac{\pi}{3}$ [5]

14. Sand falls on the horizontal ground at the rate of $9m^3$ per minute and forms a heap in the shape of a right circular cone with vertical angle 60° . Show that 10 seconds after the sand begins to fall, the rate at which the radius of the base of the pile is increasing is $\sqrt{3} \left(\frac{4}{\pi}\right)^{\frac{1}{3}} m/minute$ [8]

15. Express $\frac{3x+8}{(2x+1)(x^2+2)}$ into partial fractions [4]

THE END

DE MANUE SOLUTIONS

GENERAL CERTIFICATE OF ADVANCED LEVEL

PURE MATHEMATICS

6042/1

PAPER 1

FINAL REVISION (MONDAY 16 OCTOBER) 2023 SESSION 3 hours

Additional materials

Answer materials

Graph paper

List of formulae MF7

Non-programmable electronic calculator

Time 3 hours

INSTRUCTIONS TO CANDIDATES

Write your name in the spaces provided on the answer sheet/booklet.

Answer all questions.

If a numerical answer cannot be given exactly, and the accuracy required is not specified in the questions, then in case of angle it should be given to the nearest degree, in other cases it should be given correct to 2 significant figures.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question or part question.

The total number of marks for this paper is 120

Questions are printed in the order of their mark allocations and candidates are advised to attempt questions sequentially

The use of electronic calculator is expected where possible.

You are reminded of the need for clear presentations in your answers.

1) Find the value of $\frac{7^x + 7^{-x}}{2} = 1$ [4]

2.a) If $y = x \cos x$, find the percentage change in y when x changes from 2 to 2.01 [5]

b) The length of a cube is increasing at a rate of 5cm per second. Find the rate of change of the cube's volume when the length is equal to 20cm. [4]

3) Solve the inequality $|x + 1| > 1 - x$ [4]

4.a) find the constant in the expansion of $\left(x^2 + \frac{1}{x}\right)^9$ [4]

b.i) Expand $(p - x)^{-2}$ as a series of ascending powers of x upto and including the term in x^3 where p is a constant [5]

ii) Given that the coefficient of x^2 in the expansion is $\frac{3}{16}$, find the value of p [3]

iii) Hence state the set of values of x for which the expansion is valid [1]

5.a) Solve the equation $\cos 2\theta = \cos \theta$ for $0 \leq \theta < 2\pi$ [5]

b) Prove that $\tan \theta + \cot \theta = \sec \theta \csc \theta$ [5]

6.a) show that $\int_1^2 \frac{2x}{1+2x^2} dx = \frac{1}{2} \ln(3)$ [5]

b) Evaluate $\sum_{r=11}^{30} \frac{n}{2}(n+1)$ [4]

7.a) Find the area of the region bounded by the curve $y^2 = 4x$ and the line $y = x$ [5]

b) The region in (a) is rotated 360° about the x -axis. Find the volume generated giving your answer in terms of π [5]

8. Evaluate

$$\sum_{r=3}^6 (3r^2 + 3r - 1)$$

i) Show that $(x - 3)^2 - (x - 2)^2 \equiv 5 - 2x$

ii) Use the identity from part (b) to obtain an expression for

$$\sum_{r=1}^n (5 - 2r)$$

Hence, or otherwise, find the sum of the series $3 + 1 - 1 - 3 - 5 \dots - 35$ [15]

9. The complex number $\frac{3+2i}{2+ai}$ can be expressed in the form $x + iy$, where x and y are real. Find the value of $x = y$ [5]

10.a) A function is defined by

$$f: x \rightarrow x^2 + 4x + 1 \text{ for } x \geq -2$$

Find i) the range of the function, [2]

ii) an expression for $f^{-1}(x)$, stating the domain. [4]

11 Differentiate $f(x) = \sqrt{4x - 9}$ from first principle [5]

12. Express

$$\frac{3x + 4}{(x - 4)(x^2 - 8)}$$

in partial fractions [5]

b) hence or otherwise find the first three terms of the series expansion of

$$\frac{3x + 4}{(x - 4)(x^2 - 8)}$$

[7]

13.a) A curve has equation $4x^2 + 8x + 9y^2 - 36y + 4 = 0$

i) Find $\frac{dy}{dx}$ [4]

b) Write down the equations of the tangents of the curve that are parallel to :

i) $x - \text{axis}$ [4]

ii) $y - \text{axis}$ [4]

14. If $\frac{dy}{dx} = (xy)^2$ and when $x = 1, y = 1$, express y in terms of x [5]

15. Given that $y = \ln\left(e^{-x} - \frac{1}{2}\right)$, show that $\frac{dy}{dx} = -\frac{1}{2}(e^{-y} + 2)$ [5]

The end

DEMANUE SOLUTIONS

PURE MATHEMATICS

6042/1

PAPER 1

ZIVA PAUMIRE REVISION 2 OCTOBER 2023 SESSION

Additional material

Answer paper

Graph

List of Formulae MF7

3 hours

TIME 3 hours

INSTRUCTION TO CANDIDATES

Write your Name, Centre number and Candidate number in the space provided on the answer paper/booklet.

Answer **all** questions

If a numerical answer cannot be given exactly and accuracy is not specified in question, then in the case of an angle it should be given correct to the nearest degree, and in other cases it should be given correct to 2 significant figures.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question or part question

The total number of marks for this paper is 120

The use of non-programmable calculator is expected, where appropriate

You are reminded of the need for clear presentation in your answer.

Answer all questions

1. The remainder when $2x^3 + 2x^2 - 13x + 12$ is divided by $x + a$ is three times the remainder when it is divided by $x - a$.

a) Show that $2a^3 + a^2 - 13a + 6 = 0$. [3]

b) Solve this equation completely. [5]

2. The function f is such that $f(x) = 8 - (x - 2)^2$, for $x \in \mathbb{R}$.

a) Find the coordinates and the nature of the stationary point on the curve $y = f(x)$. [2]

The function g is such that $g(x) = 8 - (x - 2)^2$, for $k \leq x \leq 4$, where k is a constant.

b) State the smallest value of k for which g has an inverse.

For this value of k ,

c) find an expression for $g^{-1}(x)$,

d) sketch, on the same diagram, the graphs of $y = g(x)$ and $y = g^{-1}(x)$.

3. Find the equation of the graph obtained

a) when the graph of $y = x^2 + 4$ is translated -2 units in the x -axis direction.

b) when the graph of $y = x^2 + 4$ is stretched parallel to the x -axis, with y -axis invariant and a scale factor of 2. [1]

c) State precisely a sequence of geometrical transformations which would transform the graph of $y = x^2 + 4$ onto the graph of $y = -2x^2 - 6$. [3]

4a) Sketch the curve $(5x)^2 + (12y)^2 = 13^2$. Your curve should show clearly the coordinates of the intersection with the axes. [4]

b) Write down the major axis and the minor axis. [2]

5. A right circular cone has a constant volume. The height h and the base radius r can both vary. Find the rate at which h is changing with respect to r at the instant when r and h are equal. [4]

6a) Express $\cos\theta + \sqrt{3}\sin\theta$ in the form $R \cos(\theta - \alpha)$, where $R > 0$ and $0 < \alpha < \frac{\pi}{2}$, giving the exact values of R and α . [3]

b) Hence show that

$$\int_0^{\frac{\pi}{2}} \frac{1}{(\cos\theta + \sqrt{3}\sin\theta)^2} d\theta = \frac{1}{\sqrt{3}}$$

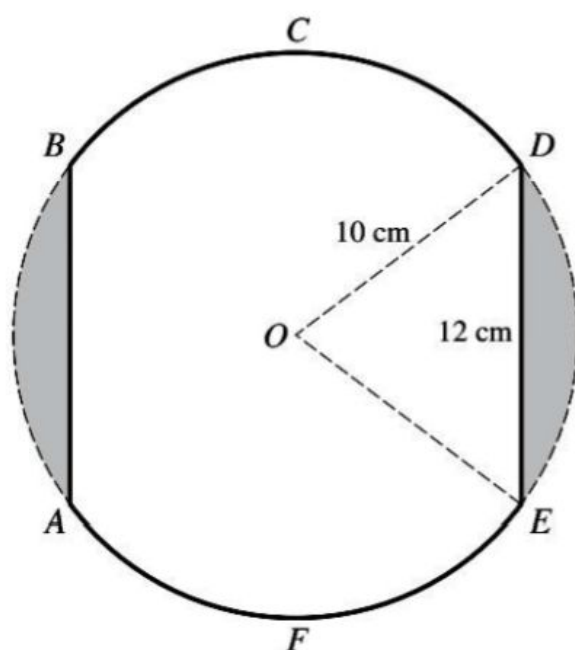
[4]

7. Given that $y = x^3 e^x$, prove by induction that, for all positive integers n ,

$$\frac{d^n y}{dx^n} = e^x [x^3 + 3nx^2 + 3n(n-1)x + n(n-1)(n-2)]. \quad [7]$$

Hence find $\int e^x (x^3 + 24x^2 + 168x + 336) dx$. [2]

8.



The diagram shows a metal plate $ABCDEF$ which has been made by removing the two shaded regions from a circle of radius 10 cm and centre O . The parallel edges AB and ED are both of length 12 cm.

a) Show that angle DOE is 1,287 radians, correct to 4 significant figures. [2]

b) Find the perimeter of the metal plate. [3]

c) Find the area of the metal plate. [3]

9a) By differentiating $\frac{1}{\cos x}$, show that if $y = \sec x$ then $\frac{dy}{dx} = \sec x \tan x$. [2]

b) Show that $\frac{1}{\sec x - \tan x} \equiv \sec x + \tan x$ [3]

c) Deduce that $\frac{1}{(\sec x - \tan x)^2} \equiv 2\sec^2 x - 1 + 2\sec x \tan x$ [2]

d) Hence show that $\int_0^{\frac{\pi}{4}} \frac{1}{(\sec x - \tan x)^2} = \frac{1}{4}(8\sqrt{2} - \pi)$ [3]

e) 10a) By sketching a suitable pair of graphs, show that the equation $2 \cot x = 1 + e^x$, where x is in radians, has only one real root in the interval $0 < x < \frac{1}{2}\pi$. [2]

b) Verify by calculation that this root lies between 0,5 and 1,0. [2]

c) Show that this root also satisfies the equation $x = \tan^{-1}\left(\frac{2}{1+e^x}\right)$ [1]

d) Use the iterative formula $x_{n+1} = \tan^{-1}\left(\frac{2}{1+e^{x_n}}\right)$ with initial value $x_1 = 0,7$, to determine this root correct to 2 decimal places. Give the result of each iteration to 4 decimal places. [3]

11a) Express $\frac{3}{(3r-1)(3r+2)}$ in partial fractions. [2]

b) Using your answers to part (a) and the method of differences, show that

$$\sum_{r=1}^n \frac{3}{(3r-1)(3r+2)} = \frac{3n}{2(3n+2)}$$
 [3]

c) Evaluate, giving your answer to 3 significant figures

$$\sum_{r=100}^{1000} \frac{3}{(3r-1)(3r+2)}$$
 [2]

Find as a single fraction

$$\sum_{r=n+1}^{3n} \frac{3}{(3r-1)(3r+2)}$$

12a) A geometric series has first term x and second term $(x^2 - x)$. If all the terms are positive, find the set of values of x for which the series converges. [3]

b) Given that $x = \frac{5}{3}$ find the smallest value of n for which the sum of the first n terms differs from the sum to infinity by less than 0,001.

13 Expand $\frac{\sqrt{1+2x}}{\sqrt[3]{1-3x}}$ in powers of x , up to and including the term in x^2 . [5]

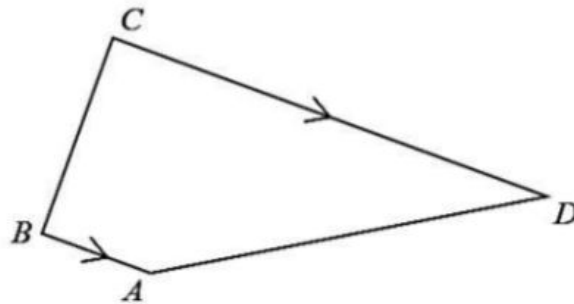
b) State the range of values of x for which the series is convergent. [1]

14a) Use Taylor series to expand $\frac{1}{x}$ as series of ascending powers of $(x - a)$, up to and

including the term in $(x - a)^3$. [5]

b) Use the expansion to evaluate $\frac{1}{1.01}$ correct to 3 significant figures. [2]

15.



The diagram shows a trapezium $ABCD$ in which BA is parallel to CD . The position vectors of A , B and C relative to an origin O are given by

$$\vec{OA} = \begin{pmatrix} 3 \\ 4 \\ 0 \end{pmatrix}, \quad \vec{OB} = \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix}, \quad \text{and} \quad \vec{OC} = \begin{pmatrix} 4 \\ 5 \\ 6 \end{pmatrix}$$

a) Use a scalar product to show that AB is perpendicular to BC . [3]

b) Given that the length of CD is 12 units, find the position vector of D . [4]

16. The complex number u is defined by $u = \frac{1+2i}{1-3i}$

a) Express u in rectangular form. [3]

b) Show on a sketch of an Argand diagram the points A , B and C representing the complex numbers u , $1 + 2i$ and $1 - 3i$ respectively. [2]

c) By considering the arguments of $1 + 2i$ and $1 - 3i$, show that

$$\tan^{-1}(2) + \tan^{-1}(3) = \frac{3}{4}\pi. \quad [3]$$

17. The variables x and y satisfy the equation $x^n y = C$, where n and C are constants.

When $x = 1,10$; $y = 5,20$ and when $x = 3,20$; $y = 1,05$.

a) Find the values of n and C . [5]

b) Explain why the graph of $\ln y$ against $\ln x$ is a straight line. [1]

End

FINAL REVISION 2: OCTOBER 2023. SESSION
0717256148.
SYNTHESIZED MARKING GUIDE

1. $f(x) = 2x^3 + 2x^2 - 13x + 12$.

$$f(-a) = 2(-a)^3 + 2(-a)^2 - 13(-a) + 12$$

$$f(-a) = -2a^3 + 2a^2 + 13a + 12$$

$$f(a) = 2a^3 + 2a^2 - 13a + 12$$

$$f(a) = 2a^3 + 2a^2 - 13a + 12$$

$$f(a) = 3f(a)$$

$$-2a^3 + 2a^2 + 13a + 12 = 3(2a^3 + 2a^2 - 13a + 12)$$

$$-2a^3 + 2a^2 + 13a + 12 = 6a^3 + 6a^2 - 39a + 36$$

$$\frac{8a^3}{4} + \frac{4a^2}{4} - \frac{52a}{4} + \frac{24}{4} = 0$$

$$2a^3 + a^2 - 13a + 6 = 0 \text{ shown.}$$

(b) Using trial and error $a-2$ is a factor

$$2a^2 + 5a - 3$$

$$a-2 \overline{) 2a^3 + a^2 - 13a + 6}$$

$$\underline{-2a^3 - 4a^2} \downarrow$$

$$5a^2 - 13a$$

$$\underline{-5a^2 - 10a} \downarrow$$

$$-3a + 6$$

$$\underline{-3a + 6}$$

$$0 \quad 0$$

$$(a-2)[2a^2 + 5a - 3] = 0$$

$$(a-2)[2a^2 + 6a - a - 3] = 0$$

$$(a-2)[2a(a+3) - 1(a+3)] = 0$$

$$(a-2)(2a-1)(a+3) = 0$$

$$a = 2; a = \frac{1}{2} \text{ or } a = -3$$

(a) Coordinates (2, 8)

Vertex $\rightarrow$ maximum turning point.

(b) $k=2$

$$c \quad y = 8 - (x-2)^2$$

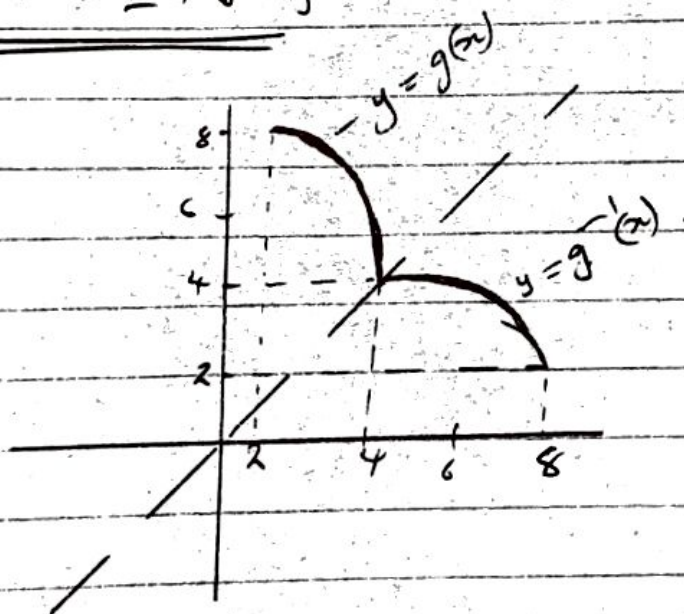
$$(x-2)^2 = 8-y$$

$$x-2 = \sqrt{8-y}$$

$$x = 2 \pm \sqrt{8-y}$$

$$\underline{\underline{g^{-1}(y) = 2 \pm \sqrt{8-y}}}$$

d.



$$2(a) \quad y = (x+2)^2 + 4$$

$$\Rightarrow x^2 + 4x + 8$$

(b)

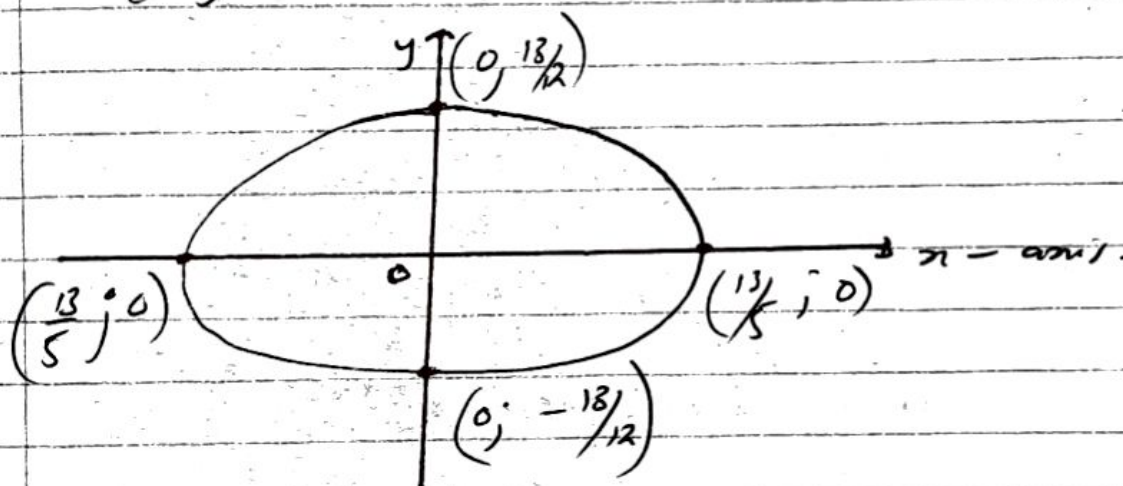
$$y = \left(\frac{x}{2}\right)^2 + 4$$

$$y = \frac{x^2}{4} + 4$$

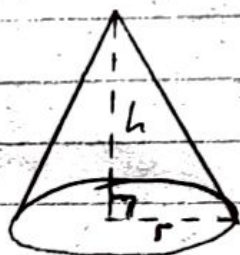
- (c) • stretch parallel to y -axis with the stretch factor of 2.
 • reflection about the x -axis
 • translation along the y -axis with the translation vector $\begin{pmatrix} 0 \\ 2 \end{pmatrix}$.

4.
$$\frac{(5x)^2}{13^2} + \frac{(12y)^2}{13^2} = \frac{13^2}{13^2}$$

$$\left(\frac{5x}{13}\right)^2 + \left(\frac{12y}{13}\right)^2 = 1$$



5.



$$V = \frac{1}{3} \pi r^2 h$$

$$\frac{dV}{dh} = \frac{2}{3} \pi r^2$$

$$\frac{dV}{dr} = \frac{2}{3} \pi r h$$

Ans: As the height increases, the radius decreases. Hence there is an inverse relationship between r and h . Therefore their gradient is negative.

$$\frac{dV}{dr} = - \frac{dh}{dV} \times \frac{dV}{dr}$$

$$= \frac{-2/3 \pi h}{\frac{1}{3} \pi r^2}$$

$$= -\frac{2h}{r}$$

When $h=r$

$$= -2 \cdot \frac{K}{K}$$

$$\underline{\underline{= -2}}$$

$$6(a) \cos \theta + \sqrt{3} \sin \theta = 2 \cos(\theta - \alpha)$$

$$R = \sqrt{1^2 + (\sqrt{3})^2}$$

$$= 2$$

$$\alpha = \tan^{-1}\left(\frac{\sqrt{3}}{1}\right)$$

$$= \pi/3$$

$$\cos \theta + \sqrt{3} \sin \theta = 2 \cos(\theta - \pi/3)$$

(6).

$$\int_0^{\pi/2} \frac{1}{[\sqrt{2} \cos(\theta - \pi/3)]^2} d\theta = \int_0^{\pi/2} \frac{1}{4 \cos^2(\theta - \pi/3)} d\theta$$

$$= \frac{1}{4} \int_0^{\pi/2} \sec^2(\theta - \pi/3) d\theta$$

$$= \frac{1}{4} \left[\tan(\theta - \pi/3) \right]_0^{\pi/2}$$

$$= \frac{1}{4} \left[\tan\left(\frac{\pi}{2} - \frac{\pi}{3}\right) - \tan\left(0 - \frac{\pi}{3}\right) \right]$$

$$= \frac{1}{4} \left[\frac{\sqrt{3}}{3} - (-\sqrt{3}) \right]$$

$$= \frac{1}{4} \left[\frac{4\sqrt{3}}{3} \right]$$

$$= \frac{\sqrt{3} \times \sqrt{3}}{3 \sqrt{3}}$$

$$= \frac{1}{\sqrt{3}} \text{ shown.}$$

Let P_n denotes the given statement.

Proof for $n=1$

$$\text{LHS} \Rightarrow \frac{dy}{dx} = e^x \cdot 3x^2 + x^3 e^x.$$

$$= e^x (x^3 + 3x^2)$$

$$\text{RHS} = e^x (x^3 + 3x^2)$$

Since LHS = RHS, P_n is true for $n=1$.

Assume true for $n=k$.

$$\frac{d^k y}{dx^k} = e^x [x^3 + 3kx^2 + 3k(k-1)x + k(k-1)(k-2)]$$

Proof for $n=k+1$.

$$\frac{d^{k+1} y}{dx^{k+1}} = e^x [x^3 + 3(k+1)x^2 + 3k(k+1)x + k(k+1)(k-1)]$$

$$\frac{d^{k+1} y}{dx^{k+1}} = \frac{d}{dx} \left[\frac{d^k y}{dx^k} \right]$$

$$= e^x [x^3 + 3kx^2 + 3k(k-1)x + k(k-1)(k-2) + e^x [3x^2 + 6kx + 3k(k-1)]]$$

$$= e^x [x^3 + 3(k+1)x^2 + 3k(k+1)x + k(k+1)(k-1)]$$

$$= e^n [x^3 + 3kx^2 + 3k(k-1)x + k(k-1)(k-2) + 3n^2 + 6kn + 3k(k-1)]$$

$$= e^n [n^3 + (3k+3)n^2 + (3k^2 - 3kn + 6k)n + k(k-1)(k-2) + 3k(k-1)]$$

$$= e^n [n^3 + 3(k+1)n^2 + 3k(k+1)n + k(k+1)(k-2+3)]$$

$$= e^n [n^3 + 3(k+1)n^2 + 3k(k+1)n + k(k+1)(k+1)]$$

P_n is true for $n=k+1$

Since P_n is true for $n=1$, $n=k$ and $n=k+1$, therefore by mathematical induction P_n is true for all positive integral values of n .

By comparing the coefficients of x^2 .

$$3n = 24$$

$$n = 8$$

$$\int \frac{d^8 y}{dn^8} = \frac{d^7 y}{dn^7} + C$$

$$= e^n (x^3 + 24x^2 + 126x + 210) + C$$

$$8(a) \quad \widehat{DOE} = 2 \sin^{-1} \left(\frac{6}{10} \right)$$

$$= 1.29 \text{ radians (3.s.f.)}$$

$$(b) \quad \text{Perimeter} = 2[10 \times (\pi - 1.29)] + 2(12)$$

$$= 61.091$$

$$= \underline{61 \text{ cm}} \quad (2.s.f.)$$

$$(c) \quad \text{Area} = 2 \left[\frac{1}{2} \times (10)^2 \times (\pi - 1.282) \right] + 2 \left[\frac{1}{2} \times (10)^2 \times \sin 1.282 \right]$$

$$= 281.465 \dots$$

$$= \underline{281 \text{ cm}^2}$$

$$9. y = \sec x \Rightarrow y = \frac{1}{\cos x}$$

$$u = 1$$

$$v = \cos x$$

$$\frac{du}{dx} = 0$$

$$\frac{dv}{dx} = -\sin x$$

Apply the quotient rule.

$$\frac{dy}{dx} = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$$

$$= \frac{(\cos x)(0) - (1)(-\sin x)}{(\cos x)^2}$$

$$= \frac{\sin x}{\cos^2 x}$$

$$= \frac{\sin x}{\cos x} \times \frac{1}{\cos x}$$

$$= \tan x \sec x$$

$$= \sec x \tan x \text{ shown.}$$

$$(a) \frac{1}{(\sec x - \tan x)^2} = (\sec x + \tan x)^2$$

$$= \sec^2 x + 2 \tan x \sec x + \tan^2 x$$

$$= \sec^2 x + \sec^2 x - 1 + 2 \tan x \sec x$$

$$= 2 \sec^2 x - 1 + 2 \sec x \tan x \text{ shown.}$$

$$(b) \frac{1}{(\sec x - \tan x)} \times \frac{\sec x + \tan x}{\sec x + \tan x} = \frac{\sec x + \tan x}{\sec^2 x - \tan^2 x}$$

$$= \frac{\sec x + \tan x}{1 + \tan^2 x - \tan^2 x} = \underline{\underline{\sec x + \tan x \text{ shown.}}}$$

$$(d). \int_0^{\pi/4} (2\sec^2 x - 1 + 2\sec x \tan x) dx.$$

$$= \left[2 \tan x - x + 2 \sec x \right]_0^{\pi/4}$$

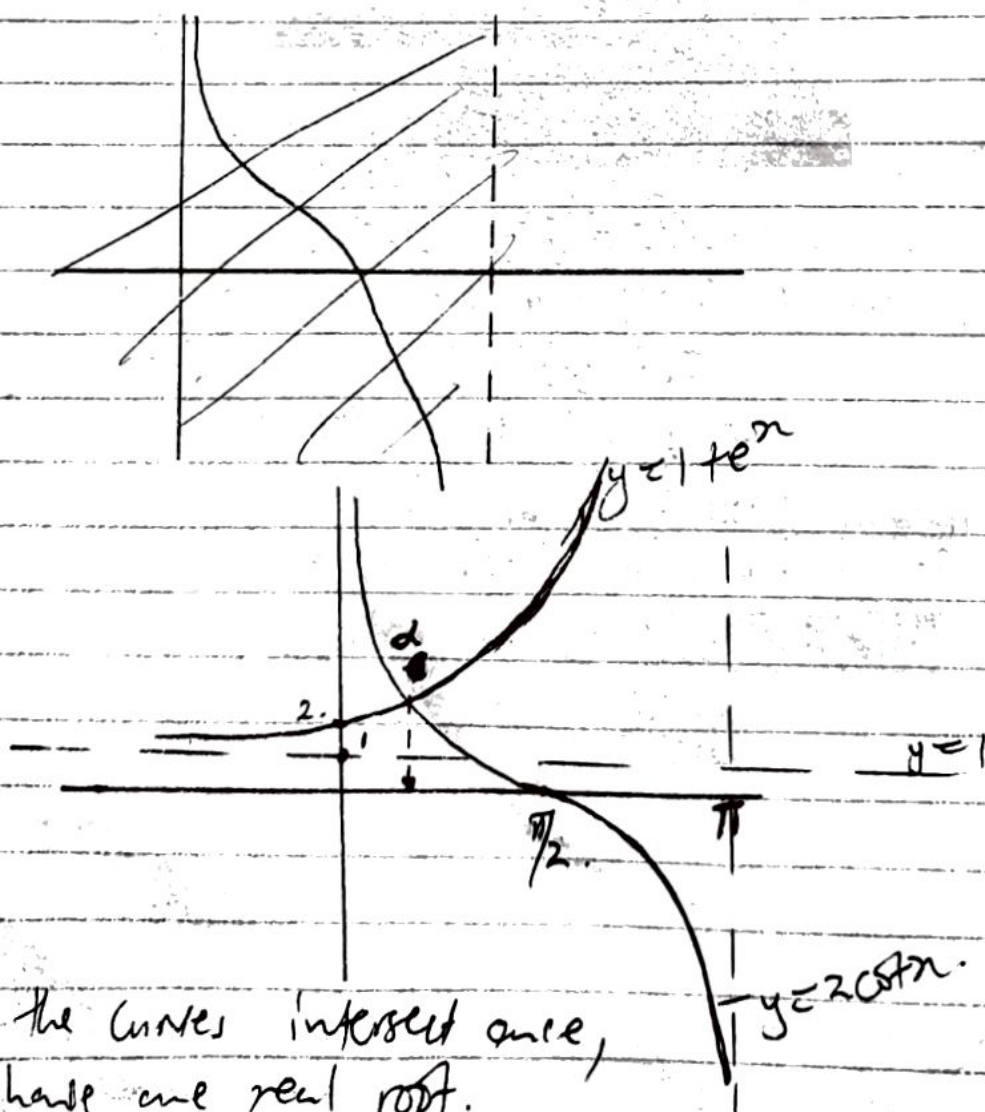
$$= \left[2 \tan x - x + \frac{2}{\cos x} \right]_0^{\pi/4}$$

$$= \left(\frac{\pi}{4} + 2 + 2\sqrt{2} \right) - (2)$$

$$= 2\sqrt{2} - \frac{\pi}{4}$$

$$= \frac{1}{4} (8\sqrt{2} - \pi) \text{ shown}$$

10(c).



Since the curves intersect once,
they have one real root.

b. $f(x) = 2 \cos x - 1 - e^x.$

$$f(0,5) = \frac{2}{\cos 0,5} - 1 - e^{0,5} = 1,012.$$

$$f(1,0) = \frac{2}{\cos(1)} - 1 - e^1 = -2,434.$$

Since there is a sign change the root lies between 0,5 and 1,0.

11(a) $\frac{3}{(3r-1)(3r+2)} = \frac{A}{3r-1} + \frac{B}{3r+2}.$

Applying cover up method.

When $r = \frac{1}{3}; A = 1$

When $r = -\frac{2}{3}; B = -1$

$$\frac{3}{(3r-1)(3r+2)} = \frac{1}{3r-1} - \frac{1}{3r+2}.$$

(b) $\sum_{r=1}^n \left(\frac{1}{3r-1} - \frac{1}{3r+2} \right)$

$r=1$	$\frac{1}{2}$	$-$	$\frac{1}{5}$
$r=2$	$\frac{1}{5}$	$-$	$\frac{1}{8}$
$r=3$	$\frac{1}{8}$	$-$	$\frac{1}{11}$
	$\frac{1}{11}$		
	$\frac{1}{14}$		
	$\frac{1}{17}$		

$$f = n-1 - \frac{1}{3n-4} - \frac{1}{3n-1}$$

$$f = n - \frac{1}{3n-1} - \frac{1}{3n+2}$$

$$= \frac{1}{2} - \frac{1}{3n+2}$$

$$= \frac{3n+2-2}{2(3n+2)}$$

$$= \frac{3n}{2(3n+2)} \quad \text{shown}$$

$$(c) \left[\frac{3n}{2(3n+2)} \right]_{99}^{1000}$$

$$= \frac{3000}{6004} - \frac{297}{598}$$

$$= \underline{0.00801354}$$

(d)

$$\sum_{r=n}^{3n} \frac{3}{(3r-1)(3r+2)} = \left[\frac{3n}{2(3n+2)} \right]_n^{3n}$$

$$= \frac{3n \times 3n}{2(3 \cdot 3n+2)} - \frac{3n \cdot n}{2(3n+2)}$$

$$= \frac{9n(3n+2) - 3n(9n+2)}{2(9n+2)(3n+2)}$$

$$= \frac{27n^2 + 18n - 27n^2 - 6n}{2(3n+2)(9n+2)}$$

$$= \frac{12n}{2(3n+2)(9n+2)}$$

$$= \frac{6n}{(3n+2)(9n+2)}$$

12(c) The sequence is a G.P.

$$a = n \quad r = \frac{n^2 - n}{n}$$

$$\frac{1}{2} = (n^2 - n)$$

$$r = \frac{n(n-1)}{n}$$

$$r = n - 1$$

for Convergent series
 $|r| < 1$

$$|n-1| < 1$$

$$-1 < n-1 < 1$$

$$0 < n \text{ or } n < 2$$

$$\underline{0 < n < 2}$$



$$(b) \quad S_{\infty} - S_n < 0.001.$$

$$\text{If } n = \frac{5}{3}; \quad a = \frac{5}{3} \quad r = \frac{2}{3}.$$

$$\frac{\frac{5}{3}}{1 - \frac{2}{3}} - \frac{\frac{5}{3} [1 - (\frac{2}{3})^n]}{1 - \frac{2}{3}} < 0.001,$$

$$5 - 5 [1 - (\frac{2}{3})^n] < 0.001.$$

$$\frac{4999}{5000} < 1 - (\frac{2}{3})^n.$$

$$(\frac{2}{3})^n < \frac{1}{5000}$$

$$\frac{n \ln(\frac{2}{3})}{\ln(\frac{2}{3})} < \frac{-\ln 5000}{\ln(\frac{2}{3})}$$

$$n > 21.01$$

$$\underline{\underline{n = 22}}$$

$$13. \quad \frac{\sqrt{1+2x}}{\sqrt{1-3x}} = (1+2x)^{\frac{1}{2}} (1-3x)^{-\frac{1}{2}}.$$

Apply Binomial Expansion.

$$(1+x)^n = 1 + nx + \frac{n(n-1)(x)^2}{2!} \dots$$

$$\cancel{1 + \frac{1}{2}(2x) + \frac{1}{2}(\frac{1}{2})(2x)^2}$$

$$= \left[1 + \frac{1}{2}(2n) + \frac{1}{2!} \left(\frac{1}{2} - 1 \right) (2n)^2 \right] \left[1 - \frac{1}{3}(-3n) + \frac{\left(\frac{1}{3} \right) \left(\frac{1}{3} - 1 \right) (-3n)^2}{2!} \right]$$

$$= \left(1 + n - \frac{1}{2}n^2 \right) \left(1 + n + 2n^2 \right)$$

$$= 1 + n + 2n^2 + n + n^2 - \frac{1}{2}n^2 + \dots$$

$$= \underline{\underline{1 + 2n + \frac{3}{2}n^2}}$$

(b) Validity of the expansion.

$$|3n| < 1$$

$$-\frac{1}{3} < n < \frac{1}{3}$$

14. (a) $f(x) = \frac{1}{x}$ $f(a) = \frac{1}{a}$

$$f'(x) = -\frac{1}{x^2} \quad f'(a) = -\frac{1}{a^2}$$

$$f''(x) = \frac{2}{x^3} \quad f''(a) = \frac{2}{a^3}$$

$$f'''(x) = -\frac{6}{x^4} \quad f'''(a) = -\frac{6}{a^4}$$

$$= \frac{1}{a} - \frac{1}{a^2}(x-a) + \frac{\left(\frac{2}{a^3} \right) (x-a)^2}{2!} + \frac{\left(-\frac{6}{a^4} \right) (x-a)^3}{3!}$$

$$= \underline{\underline{\frac{1}{a} - \frac{1}{a^2}(x-a) + \frac{1}{a^3}(x-a)^2 - \frac{1}{a^4}(x-a)^3}}$$

$$= \left[1 + \frac{1}{2}(2x) + \frac{1}{2!} \left(\frac{1}{2} - 1 \right) (2x)^2 \right] \left[1 - \frac{1}{3}(-3x) + \frac{\left(\frac{1}{3} \right) \left(\frac{1}{3} - 1 \right) (-3x)^2}{2!} \right]$$

$$= (1 + x - \frac{1}{2}x^2)(1 + x + 2x^2)$$

$$= 1 + x + 2x^2 + x + x^2 - \frac{1}{2}x^2 + \dots$$

$$= \underline{1 + 2x + \frac{3}{2}x^2}$$

(b) validity of the expansion.

$$\begin{aligned} |3x| < 1 \\ -\frac{1}{3} < x < \frac{1}{3} \end{aligned}$$

14) $f(x) = \frac{1}{x}$ $f(a) = \frac{1}{a}$

$$f'(x) = -\frac{1}{x^2} \quad f'(a) = -\frac{1}{a^2}$$

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$$= \frac{1}{a} - \frac{1}{a^2}(x-a) + \frac{\left(\frac{2}{a^3} \right) (x-a)^2}{2!} + \frac{\left(-\frac{6}{a^4} \right) (x-a)^3}{3!}$$

$$= \underline{\underline{\frac{1}{a} - \frac{1}{a^2}(x-a) + \frac{1}{a^3}(x-a)^2 - \frac{1}{a^4}(x-a)^3}}$$

See marks 477256148

(b) Putting $x = 1, 01$; centered @ $x = -1$

$$\frac{1}{1, 01} = 1 - (1, 01 - 1) + (1, 01 - 1)^2 - (1, 01 - 1)^3$$

$$= 0, 990 \quad 3. sf.$$

$$\text{15(a)} \quad \vec{OB} = \vec{OB} - \vec{OA}$$

$$= \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix} - \begin{pmatrix} 3 \\ 4 \\ 0 \end{pmatrix} = \begin{pmatrix} -2 \\ -1 \\ 2 \end{pmatrix}$$

$$\vec{BC} = \vec{OC} - \vec{OB}$$

$$= \begin{pmatrix} 4 \\ 5 \\ 6 \end{pmatrix} - \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix}$$

$$= \begin{pmatrix} 3 \\ 2 \\ 4 \end{pmatrix}$$

$$\vec{AB} \cdot \vec{BC} = \begin{pmatrix} -2 \\ -1 \\ 2 \end{pmatrix} \cdot \begin{pmatrix} 3 \\ 2 \\ 4 \end{pmatrix}$$

$$= -6 - 2 + 8$$

$$= 0. \text{ shown.}$$

(b). Unit vectors of the parallel vectors are equal.

$$\frac{\vec{CD}}{|\vec{CD}|} = \frac{\vec{BA}}{|\vec{BA}|}$$

$$\frac{\vec{CD}}{|\vec{CD}|} = \frac{\vec{BA}}{|\vec{BA}|}$$

$$\frac{\vec{CD}}{12} = \frac{2i + j - 2k}{\sqrt{2^2 + 1^2 + (-2)^2}}$$

DB ~ matrix of 1756148

$$\vec{OD} - \vec{OC} = \frac{12}{5} \begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix}$$

$$D = \begin{pmatrix} 8 \\ 4 \\ 8 \end{pmatrix} + \begin{pmatrix} 4 \\ 5 \\ 6 \end{pmatrix}$$

$$\vec{OD} = \begin{pmatrix} 12 \\ 9 \\ -2 \end{pmatrix}$$

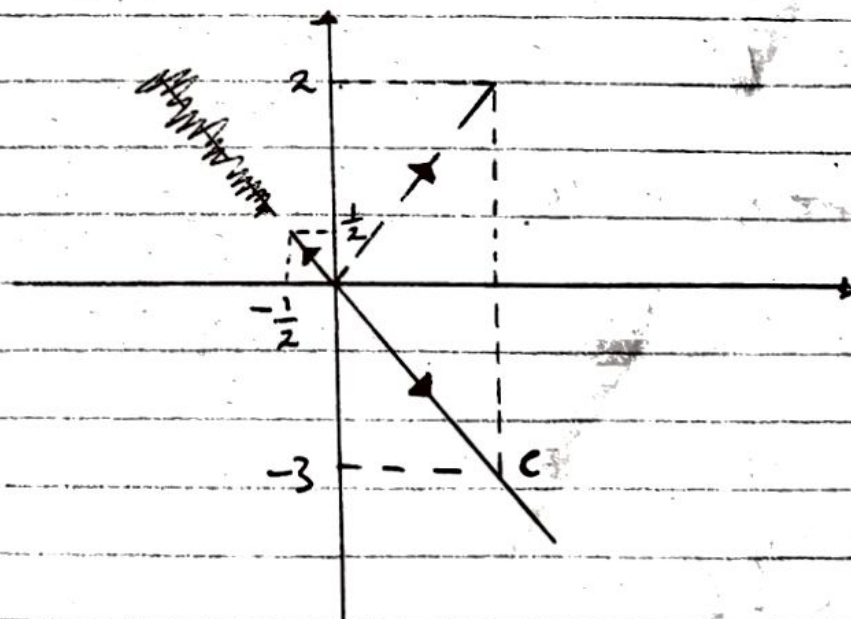
$$(ka) \quad u = \frac{1+2i}{1-3i} \times \frac{1+3i}{1+3i}$$

$$u = \frac{1+5i-6}{10}$$

$$u = \frac{-5+5i}{10}$$

$$u = -\frac{1}{2} + \frac{1}{2}i$$

(b)



$$(c) \arg\left(\frac{1+2i}{1-i}\right) = \arg\left(-\frac{1}{2} + \frac{1}{2}i\right)$$

$$\tan^{-1}(2) - \tan^{-1}(-3) = \tan^{-1}\left(\frac{\frac{1}{2}}{-\frac{1}{2}}\right) + \pi$$

$$\tan^{-1}(2) + \tan^{-1}(3) = -\pi/4 + \pi$$

$$\tan^{-1}(2) + \tan^{-1}(3) = \underline{\underline{\pi/4 \text{ shown.}}}$$

$$17) x^n y = c$$

$$\ln(x^n y) = c$$

$$n \ln x + \ln y = \ln c$$

$$\ln y = -n \ln x + \ln c$$

$$-n = \frac{\ln 1.05 - \ln 5.20}{\ln 3.20 - \ln 1.10}$$

$$-n = -1.498$$

$$\underline{\underline{n = 1.5 \text{ 2.s.f.}}}$$

$$\ln 5.20 = -1.5 \ln 1.10 + \ln c$$

$$\ln 5.20 + 1.5 \ln 1.10 = \ln c$$

$$\ln(5.20 \times 1.10^{1.5}) = \ln c$$

$$5.20 \times 1.1^{1.5} = c$$

$$c = 5.999$$

$$\underline{\underline{c = 6.0}}$$

- (b) Because it has linear equation which is in the standard form $y = mx + c$.

THE END.

Compiled by Emmanuel Zisanayi

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